

JetFile II Protocol Ver2.5.4

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Protocol update Versions Record

Protocol Version No.:	Protocol Update Contents & Cause (What and Why)	Updated by	Date Updated
Ver2.1	1. Changed to use Data length to record the length of the following data 2. Changed to use the first 2B as the source address. In the second 2B, the first byte means (GG) and the second byte means (UU). Both GG & UU are of decimal denotation and of not ASCII. 3. Changed the Packet Serial numbers from 4 bits to 2 bits 4. Outgoing data prefix is 0x55 0xa7 5. Returning data prefix is 0x55 0xa8 6. Changed the parameters length from 2 bytes to 1 byte. The last byte is reserved for future use 7. The format of feedback data & sending data is the same. The first 16 outgoing bytes are the same as the received data. The others bytes are decided by actual application. 8. There are two formats: return data format & error feedback format. If the data is received successfully, the user should receive feedback to make sure bilateral data communication 9. In reading system configuration parameters, there are 16 bytes. Packet size data and packet serial numbers are all put into parameters. But the data segment is left blank. 10. The file name is changed to 12 bytes providing room for future long file names.	Mr. Yang Lei	DEC. 30.2005
Ver2.1.1	Use the Rev. data files as flag to mark whether feedback data is needed or not.	Mr. Yang Lei	DEC. 30.2005
Ver2.1.2	1. The data feedback codes table is changed. 2. Changed to ignore the flag mark in reading information command major category. Because every sending data would have feedback. 3. Moved the command of reading a file on a designated path after the array picture file reading commands. Moved the file write-in to designated path command to after the array picture file writing commands.	Mr. Yang Lei	JAN. 3.2006
Ver2.1.3	1. Make the F Disk invalid for commands. Because F Disk commands may cause many unreliable problems and the user can only use other default sections for operation. 2. The data feedback codes table is updated	Mr. Yang Lei	JAN. 11. 2006
Ver2.1.4	1. Add the command for non word-wrap display status commands. These are the so-called Category IX commands. 2. Change on the format of feedback of system parameter reading	Mr. Yang Lei	JAN. 13. 2006

	commands. The last two bytes are reserved for GGUU .		
Ver2.1.5	Add a mark before display log file	Mr. Yang Lei	JAN. 16.2006
Ver2.1.6	1. Changed the return codes for unlimited connection display. Changed the meanings of the two return codes (feedback codes) 8301 & 8302. And the return code 8303 is added. All return codes can tell which two buffs are being displayed in the system. 2. Change the unlimited connection display return codes	Mr. Yang Lei	JAN. 17. 2006
Ver2.1.7	1. Delete error code 9003. When the address is wrong, no return data is given. 2. Add the non word-wrap status to the Command Quick Check Sheet	Mr. Yang Lei	JAN. 18. 2006
Ver2.1.8	Revision 1: Two commands are added: 1. One command to read SN, MAC structure body (0x1000) 2. One command to write SN, MAC structure body (0x2000) Revision 2: One command is added to the first communication format. This command is to delete all files. The command is E\$.	Mr. Yang Lei	FEB.06. 2006
Ver2.1.9	1. Command 020c was deleted. Configuration system initial status was placed in configuration files. 2. Some ADP compatible commands were added (to enhance the first communication format) 3. Corrected some spelling errors and misleading expressions. 4. Modified “open connection test” command, and added an option to allow selection of data format for sending. 5. Added a status (return code)	YL	APR.01.2006
Ver2.1.10	1. Length of parameter in read command return format changed from 2 bytes to 4 bytes. 2. One command DIR(070c) was added to allow reading by package. 3. Partitioning code in deleting command changed to two modes(1,2,3...or C,D,E...)	YL	APR.14.2006
Ver2.1.11	1. Two commands were added in play control, being 1)read current play file name and contents by package, and 2) read next play file name and contents by package. 2. The following error codes were added: 0x7201(Formatting failed) 0x9012(file writing failed) 0x2102(Space on disk not enough) 0x7401(Renaming file failed) 0x7402(Wrong path in renaming) 0x6702(failure in opening a display file) 0x6703(Current file oversize. Please use extended reading command) 0x9030(Please log in first. If password is required but operation is attempted without logging in, the software will return back).	YL	APR.26.2006

	0x9031(Password not correct). 0x9032(User name not correct) 0x903(Old password not correct) 0x9034(Password has been changed successfully) 0x7D01 //Disk reading failed (Disk not existing or wrong disk name) 3. A restriction was set in writing by package. Packages have to be sent by order. 4. A major category command 0x0A was added for login. 5. A file control command 070D is added for reading disk information. 6. Modified “open connection test” command, and added an option to allow format for sending data(meaning of revolving values 1 and 0). 7. A command (010D) for reading Flash writing status was added.		
Ver2.1.12	1. A command was added for reading cabinet work status. 2. A return error code (0x9035) was added.	YL	MAY.23.2006
Ver2.2	1. Add one CMD to decide update whether success 2. ADD repose code	YL	JUN.03.2006
Ver2.2.1	ADD one CMD for read humidity.	SOAR	JUN.17.2006
Ver2.3	Modify first one communication format	SOAR	JUL.07.2006
Ver2.3.1	Delete something for self	SOAR	JUL.25.2006
Ver2.3.2	1. modify “Brightness Control Block Write-in(0x020a)” CMD’s Arg 2. modify “Reading Current System Status Command(0x010B)”CMD’s 3. Arg. Add humidity 4. Reading System Arg. Command(0x010A) [2 bytes]type →[2 bytes];protocol version 5. delete cmd E\$A 6. modify cmd read/write to Designated Path E.g. To ”D:\T\WELCOME.NMG”. 7. Verify Table2.7.2 Default Value fields W and V. 8. Two reading and writing default commands were added(0x0110and 0x020C)5	SOAR	AUG.22.2006
Ver2.3.3	Two commands were added for reading large files(>80M). One command is to read designated path extension.	SOAR	OCT.31.2006
Ver2.3.4	A command was added to read system information.	SOAR	NOV.27.2006
Ver2.3.5	1. Modified log file recording format. 2. A pair of commands were added for switching ON and OFF the sign.	SOAR	DEC.30.2006
Ver2.3.6	A command was added to read power-off status.	SOAR	JAN.10.2007
Ver2.3.7	A command was added to test gray scale.	SOAR	JAN.17.2007
Ver2.3.8	Command 0119 was added for reading system error log.	SOAR	JAN.22.2007
Ver2.3.9	1. A control buzzer command was added to play control. 2. In protocol, remarks were given about which-for-which among various kinds of control boards. 3. Appendix XI was added glossary and definitions. All appendices were updated.	SOAR	FEB.10.2007

Ver2.3.10	control buzzer command was be modified	TONY	APR.03.2007
Ver2.4.0	Correct the language mistakes in Ver2.3.10	SOAR	MAY.14.2007
Ver2.5	Modify: combine the file format and the display protocol into this file.	SOAR	JUN.09.2007
Ver2.5.1	Correct the language mistakes	SOAR	JUN.25.2007
Ver2.5.2	Correct the arrange mistakes	SOAR	JUN.25.2007
Ver2.5.3	Correct the “sequent.sys” file discription	SOAR	AUG.14.2007
Ver2.5.4	Add command 0x0504(Time Adjust)	SOAR	2007-11-2

Part I: Communicating Protocol

I、Outline

Brief introduction

There are two kinds of communicating formats in JetFileII: Compared with the 2nd communicating format, **the 1st communicating format is simple and suitable for some basic industry applications. The 2nd communicating format is more complicated, powerful and systematic.**

Comparison of the two formats:

- 1st communicating format:
1. Support the written-in of Text file, String file, Picture file, Array picture file;
 2. No read-back command;
 3. No delete designated file command
 4. Maximum length of the file name: 2 characters
 5. Do not support dividing package to send;
 6. No corresponding checksum;
 7. Support the display mode of mission scheduling only

- 2nd communicating format:
1. Support all written-in and read-back of Text File, String file, Picture file, Array picture, Fonts, Configuration,etc.
 2. Support to delete of any file;
 3. Support dividing package to send (it can send more than 1024 bytes file);
 4. Corresponding checksum (Enhancing the communication);
 - 5.Support multi-displaying modes (such as mission scheduling, Non word-wrap display and so on)

This protocol can be based on the communication of TCP/IP, UDP/IP, COM, GPS etc.

Explanation of the file format:

1. **<0x> represents that the value of the character is hexadecimal. “ / ” those enclosed in bracket are character or string, value of the rest is decimal.**
2. **Unless special remarks were given, all commands in the files are available for QS0712, QS5006, QS5003 and QS0925.**
3. **The data rank in the file is in Little End Format: it means what is saved in low address is the low byte of character data, what is saved in high address is the high byte of character data. For example, the rank of 0x11223344 in the memory is:**

0			3
44	33	22	11

4. **In this system, File name in upper and lower case are differentiated, but not for path or drive.**

1、The 1st communicating format

This format is mainly used for saving Text file, String file, Picture file, Array picture file to the related catalogue. The format is shown in the following Table1.1:

Table1.1 1st communicating format

<SOH>	<REV>	Sign Address	<STX>	Command Code	File Label	Data Field	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, <SOH> value =0x01				
<REV>	1 Byte	1	Fixed. Fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start address of the command. All the characters behind this character are the Command code. <STX> value=0x02				
Command Code	1 Byte	5	Command code				
File Label	1 Byte	6	File name. Please refer to the file name list for further information.				
Data Field	N Bytes	7	The size is determined by the data. Please note that the size can't exceed 1024 Bytes				
<EOT>	1 Byte	7+data	End command , <EOT> value =0x04 or 0x03 0x04: In-echo, 0x03: echo				

2、The 1st echo communicating format

Echo data format after successful operation (if required)

Echo character		
Name	Size	Description
Echo data	2 Bytes	Value of successful operation: 'OK' The related status code will be echoed if the operation fails (For detailed info please refer to the explanation in the 1 st communicating format.)

3、The 2nd communicating format.

This format is used for complicated data communication. Table1.2 explains the format of this type.

Table1.2 2nd communicating format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	Main CMD	Sub CMD	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Name	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Synchronic character. Value=0x55 0xa7
B	Check Sum	2	2 Bytes	Calculation: Starts from the first byte of the sum-check and ends at the last byte. The value is saved under WORD format while the part that exceeds WORD will be delete.(These value does not have the symbol calculation) , Calculation function can be found in Appendix.
C	Data Len	4	2 Bytes	Length of the data, which determines the length of the transforming data.
D	Source Address	6	2 Bytes	Source address
E	Destination address	8	2Bytes	The pre-data represent the Group address, while the post-data represent the Unit address. Value scope is [0-254], There are 254 addresses in total and 00 is the broadcast address. GG: Group address; UU: Unit address.
F	Packet Serial	10	2 Bytes	The serial number of each packet recognizes an unique packet during the same period
G	Main CMD	12	1 Byte	Main command Type
H	Sub CMD	13	1 Byte	Subordinate command type
I	Arg.Len	14	1 Byte	The Arg. field length is 4 times as the value of the Arg.Len. E.g.: When value=1, the Arg. field length=1 * 4 Bytes When value=2, the Arg. field length=2 * 4 Bytes
J	Flag	15	1 Byte	1 = Echo 0 = In-echo
K	Arg.	16	4*N Bytes	It is determined by the length of the Arg.Len field. Empty value is allowed
L	Data	16+N	N Bytes	Transforming data. The definition varies with the command. The length is decided by the Data Len. Empty value is allowed..

4、The 2nd echo communicating format.

Format of the 2nd echo communicating format (if required for echoing or reading back)

Table1.3 General Communication Echo Data Format.

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	Main CMD	Sub CMD	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L
Name	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Synchronic data. Value=0x55 0xa8							
B	Checksum	2	2 Bytes	Calculation: Starts from the first byte of the sum-check and ends at the last byte. The value is saved under WORD format while the part that exceeds WORD will be delete.(These value does not have the symbol calculation)							
C	Data Len	4	2 Bytes	Length of the data, which determines the length of the transforming data.							
D	Source Address	6	2 Bytes	Source address							
E	Destination address	8	2Bytes	The pre-data represent the Group address, while the post-data represent the Unit address. Value scope is [0-254], There are 254 addresses in total and 0 is the broadcast address. GG: Group address; UU: Unit addresses.							
F	Packet Serial	10	2 Bytes	Same as the received packet serial number, and showed which packet is echoed							
G	Main CMD	12	1 Byte	Same as the received main command type							
H	Sub CMD	13	1 Byte	Same as the subordinate command type							
I	Arg.Len	14	1 Byte	The Arg. field length is 4 times as the value of the Arg.Len. E.g.: When value=1, the Arg. field length:1 * 4 Bytes When value=2, the Arg. field length:2 * 4 Bytes							
J	Flag	15	1 Byte	If the value = 1, it is the echo code, If the value = 0, it is the data.							
K	Arg.	16	4*N Bytes	The length is determined by the types. Empty value is allowed							
L	Data	16+N	N Bytes	Transforming data. The definition varies with the command. The length is decided by the Data Len. Empty value is allowed..							

5、 Examples of the two formats:

Send a Text File named '0' to the default disk. Its content is "This is a sample".

The 1st communicating format:

<0x01>Z00<0x02>A0<0x06>This is a sample<0x04>

Meaning:

<0x01>: Start symbol of the command

Z : Fixed

00 : Broadcast address

<0x02>: Start address of the command

A :Command for write in Text File

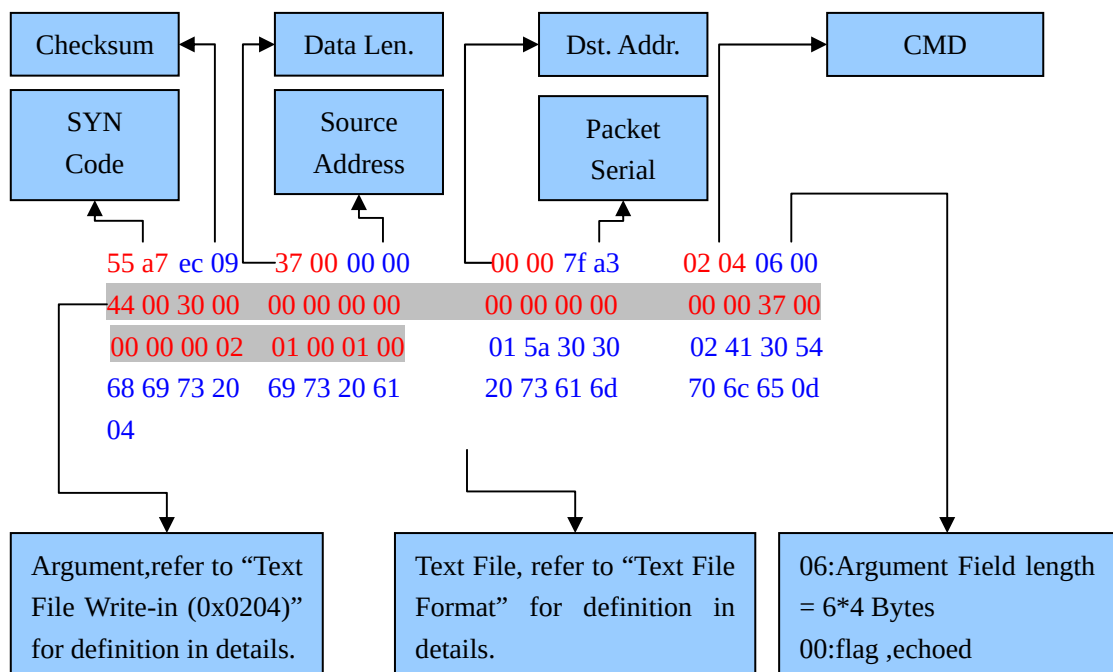
0 :File label

<0x06>: If without this control character, under the 1st communication of Jetfile II display protocol, the system will use the ADP2.0 display protocol.

This is a sample: Content

<0x04>: End command, In-echo.

The 2nd communicating format: communicating data as bellow, which all are hexadecimal



II 、 Explanation of the 1st communicating

1、 Write in Text File

It is used for writing the TEXT FILE to the default disk or the T catalogue in the designated disk. The format as the following table:

Table2.1.1 Command format of writing in Text File

<SOH>	<REV>	Sign Address	<STX>	Command Code	File Label	Data Field	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, <SOH> value=0x01				
<REV>	1 Byte	1	Fixed. Fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, value = 'A'				
File Label	1-5Bytes	6	File name 1.If the 1 st character $\neq$ <0x0F>, this filed will have one file name only and be saved in the default disk. 2.If the 1 st data=<0x0F>, it represents a path, with the format:<0x0F><Disk Number><Catalogue><File Name(2 Bytes long)> E.g.:<0x0F>DTT1 D --- D drive T --- T folder T1 --- File name				
Data Field	N Bytes	7/11	Text File data. Please note that the size can't exceed 1024 Bytes Note:This Text File only have data field ,no File head field ,no EOF field.,refer to Text File format for more information.				
<EOT>	1 Byte	7/11+N	End code 1. If Value=0x04, In-echo 2. If Value=0x03, Echo.				

Note of Echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as 'OK'. Status code will appear when the operation fails.

2、 Write in String File

It is used to write the String File to the default disk or the S catalogue in the designated disk.
The format as the following table:

Table2.2.1 Command Format of writing in String File

<SOH>	<REV>	Sign Address	<STX>	Command Code	File Label	Data Field	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, <SOH> value=0x01				
<REV>	1 Byte	1	Fixed. Fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, Value='G'				
File Label	1-5Byte	6	File name 1.If the 1 st character $\neq$ <0x0F>, this filed will have one file name only and be saved in the default disk. 2.If the 1 st character =<0x0F>, it represents a path, with the format: <0x0F><Disk Number><Catalogue><File Name(2 Bytes long)> E.g.:<0x0F> DST1 D --- D drive S --- S folder T1 --- File name				
Data Field	N Bytes	7/11	String File data. Please note that the size can't exceed 1024 Bytes				
<EOT>	1 Byte	7/11+N	End code 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.				

Special usage:

Send several String File under one command and use<0x02> as the partition:

<0x01>Z00<0x02>GaThis is string file a<0x02>G<0x0F>ESaaThis is string file aa<0x02>G<0x0F>ESVgThis is string file Vg<0x02>GcThis is string file c<0x04>

The above command equals to the following 4 commands:

<0x01>Z00<0x02>GaThis is string file a<0x04>
 <0x01>Z00<0x02>G<0x0F>ESaaThis is string file aa<0x04>
 <0x01>Z00<0x02>G<0x0F>ESVgThis is string file Vg<0x04>
 <0x01>Z00<0x02>GcThis is string file c<0x04>

Note of Echo: when the <EOT> value= 0x04, no data will echo back. When <EOT> value =0x03 and operate successfully, data echoes as 'OK'. Status code will appear when the operation fails.

3、 Writing in Picture File

It is used to write the Picture File to the default disk or the P catalogue under the designated disk. This command does not support packet dispatching. Only the small pictures can be sent. If large pictures are used, please use the 2nd communicating format. The format as the following table:

Table2.3.1 Command format of writing in Picture File

<SOH>	<REV>	Sign Address	<STX>	Command Code	File Label	Data Field	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, <SOH> value=0x01				
<REV>	1 Byte	1	Fixed. Fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, value='I'				
File Label	1-5Bytes	6	File name 1.If the 1 st character $\neq$ <0x0F>, this filed will have one file name only and be saved in the default disk. 2.If the 1 st character = <0x0F>, it represents a path, with the format: <0x0F><Disk Number><Catalogue><File Name(2 Bytes long)> E.g.:<0x0F> DPT1 D --- D drive P --- P folder T1 --- File name				
Data Field	N Bytes	7/11	Picture File data. Please note that the size can't exceed 1024 Bytes				
<EOT>	1 Byte	7/11+N	End code 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.				

Note of Echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as 'OK'. Status code will appear when the operation fails.

4、 Writing in Array Picture File

It is used to write the Picture File to the default disk or the A catalogue under the designated disk. Since this command doesn't support sending partial packets, user can send some small pictures, and if user wants to send big pictures, please use the second communication format. The format as the following table:

Table2.4.1 Command format of writing in Array Picture File

<SOH>	<REV>	Sign Address	<STX>	Command Code	File Label	Data Field	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, <SOH>value is =0x01				
<REV>	1 Byte	1	Fixed, fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, the value ='K'				
File Label	1-5Bytes	6	File name 1.If the 1 st character $\neq$ <0x0F>, this filed will have one file name only and be saved in the default disk. 2.If the 1 st data=<0x0F>, it represents a path, with the format:<0x0F><Disk Number><Catalogue><File Name(2 Bytes long)> E.g.:<0x0F> DAT1 D --- D drive A --- A folder T1 --- File name				
Data Field	N Bytes	7/11	Array picture Data, the size is decided according to the actual data. but the size of data + the size of protocol packet can not exceed 1024 Bytes				
<EOT>	1 Byte	7/11+N	End code 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.				

Note of Echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as 'OK'. Status code will appear when the operation fails.

5、Operation of Deleting the message

This command is used to delete all the Text Files and all the String Files etc. But it doesn't support deleting file by name. If the user wants to delete the designated file, please use the second communication format. The format as the following table:

Table2.5.1 Command format of deleting

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	<EOT>
Name	Size	Offset	Description			
<SOH>	1 Byte	0	Start symbol of the command, <SOH> value is= 0x01			
<REV>	1 Byte	1	Fixed, fill in as ‘Z’			
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.			
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02			
Command Code	1 Byte	5	Command code, the value is=‘E’			
Arg.	1-2 Bytes	6	‘S’	‘T’	Delete Text File	
				‘S’	Delete String File	
				‘P’	Delete Picture File	
				‘Q’	Delete Play List	
				‘R’	Delete Run time table	
				‘A’	Delete all	
				If there is no this field, it also means deleting all		
<EOT>	1 Bytes	7/8	Symbol of the end of a command 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.			

Note of Echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as 'OK'. Status code will appear when the operation fails.

6、Operation of playing the list

It is used to operate the Play List. The format as the following table:

Table2.6.1 Command format of the playing List

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	File List	<EOT>
Name	Size	Offset	Description				

<SOH>	1 Byte	0	Start symbol of the command, the value of<SOH>=0x01
<REV>	1 Byte	1	Fixed, fill in ‘Z’
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02
Command Code	1 Byte	5	Command code, the value is: ‘E’
Arg.	3 Byte	6	The value = “.SL”
File List	N Byte	9	File name List If the file name is<0x0F>, it means the following is a path. E.g.:02<0x0F>DTAB1, it means the following list: 1. File named 0 in Default disk, 2. File named 2 in Default disk, 3. D:\T\AB, 4. File named 1 in Default disk.
<EOT>	1 Byte	9+N	Ending symbol of Command 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.

Note for special usage(deleting Play List):

Table2.6.2 Command format of deleting playing List

<SOH>	<REV>	Sign Address	<STX>	Command Code	<EOT>
Name	Size	Offset	Description		
<SOH>	1 Byte	0	Start symbol of the command, the value of<SOH>=0x01		
<REV>	1 Byte	1	Fixed, fill in ‘Z’		
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.		
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02		
Command Code	2 Byte	5	Command code, the value is: “.E.”		
<EOT>	1 Byte	7	Ending symbol of Command 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.		

For example, the following command means deleting the Play List:

<0x01>Z00<0x02>E.<0x04>

7、Set the default display mode

It is used to setting the default display parameter of the LED sign, E.g.: setting the size of the default characters, if the control characters are not used in font selection in the Text File, the default character will be used. The format as the following table:

Table2.7.1 Command format of setting the default display mode

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	Default Value	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, the value of <SOH> =0x01				
<REV>	1 Byte	1	Fixed, fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, the value = 'E'				
Arg.	1 Byte	6	The value is '#'				
Default Value	N Bytes	7	List of default display modes, please refer to Table2.7.2 for more meanings				
<EOT>	1 Byte	7+N	The ending symbol of a command 1.If Value=0x04, In-echo 2.If Value=0x03, Echo.				

Table2.7.2 Explanation of Default Value field

Default Value Field Format:	
Default Variable	Value
Default Variable	Value
'B'	Background color '1' = Red '2' = Green '3' = Yellow '4' = Black (Factory default)
'C'	Primary color '1' = Red (Factory default) '2' = Green '3' = Yellow '4' = Black
'D'	Default Drive 'E' = RAM (random access drive – volatile) 'D' = Flash (non-volatile) memory (Factory default) 'F' = Nor Flash (non-volatile) or CFCARD memory
'H'	Horizontal Justification 'C' = Center 'T' = Top 'B' = Bottom 'F' = Fill (Factory default)
'L'	Line spacing '1' = 1 pixel (Factory default) '2' = 2 pixel '3' = 3 pixel

	'4' = 4 pixel '5' = 5 pixel ETC.	
'F'	Default Font '0' = Normal 5 '1' = Normal 7 (Factory default) '2' = Normal 14 '3' = Normal 15 '4' = Normal 16 '5' = Bold 14 '6' = Bold 15 '7' = Bold 16	
'M'/'O'	In Mode / Out Mode	
	'a' = rotate 'b' = hold 'c' = flash 'd' = random(Factory default) 'e' = move left 'f' = move right 'g' = move up 'h' = move down 'I' = scroll o/c 'j' = scroll o/r 'k' = scroll o/l 'l' = unveil in 'm' = unveil up/in 'n' = unveil up 'o' = unveil down 'p' = unveil up/out 'q' = splice across 'r' = splice verti 's' = fall left 't' = fall right 'u' = veretian hor 'v' = veretian ver 'w' = rain 'x' = materialize 'z' = twinkle 'Q' = rit2lftupcor 'R' = grow up	'1' = squigle '2' = radar '3' = fan open '4' = fan close '5' = rotate right '6' = rotate left '7' = center2corner '8' = corner2center '9' = center2allsz 'A' = alls2center 'B' = fourblock2cor 'C' = fourblock2cen 'D' = fourblockout 'E' = fourblockin 'F' = leftcorrectin 'G' = rightcorrectin 'H' = lbottomrecti 'I' = rbottomrecti 'J' = lftcodiagonal 'K' = rtcodiagonal 'L' = lbtmdiagonal 'M' = rbtmdiagonal 'N' = lft2rtdowncor 'O' = rt2lftdowncor 'P' = left2rtupcor
'S'	Presentation Speed '1' = Slowest Speed '2' = Slow Speed '3' = Medium Slow Speed '4' = Medium Speed '5' = Medium Fast Speed (Factory default) '6' = Fast Speed '7' = Fastest Speed	
'T'	Time for the Message to stay on the display 0-9 in Seconds 3 seconds is Factory default. E.g. '99' = 99 second	
'W'	Word-wrap mode '0' = OFF Text will move across face from left to right '1' = On (Factory default)	

‘V’	Vertical Justification ‘R’= Right ‘L’= Left ‘C’= Center (Factory default)
------------	--

Note of Echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as ‘OK’. Status code will appear when the operation fails.

Note for special usage:

If there are no value in Default value field, it means the settings of factory is renewed, for example:

1. Set the default display mode: the character color is red; the default disk is E; the default character is normal 7, the in-mode and out-mode are “Jump Out”. The command is as follow:

<0x01>Z01<0x02>E#C1DEF1MbOb<0x04>

2. Renew the settings of factory, the command is as follow:

<0x01>Z01<0x02>E# <0x04>

The default value of factory is:

Font Color: RED	Back Ground Color : Black	Font Size : Normal7.FNT(7x5)
Line Space : 1 pixel	Stay Time : 3 seconds	Speed : 3rd speed
Horizontal Justify : Center	Vertical Justify: Fill	Word-wrap : On
In Mode : Random	Out mode : Random	Default Drive: D:

8、Read time | Set time

(1)Read back the time

It is used to read back the date, time and week of the system. The format as the following table:

Table2.8.1.1 Command format of Reading Time

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	<EOT>
Name	Size	Offset	Description			
<SOH>	1 Byte	0	Start symbol of the command, the value of <SOH>=0x01			
<REV>	1 Byte	1	Fixed, fill in as ‘Z’			
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.			
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02			
Command Code	1 Byte	5	Command code, the value = ‘E’			
Arg.	1 Byte	6	The value= ‘ A ’			

<EOT>	1 Byte	7	The ending symbol of a command, 0x04
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✚ **Note of echo:** If the command is carried out successfully, the data will be echoed. And its format is as the following Table2.8.1.2.

Table2.8.1.2 Echoed data format of Reading Time command

SYN Head	Return Size	Year	Month	Day	Hour	Minute	Week	Time Zone
Name	Size	Description						
SYN head	2 Bytes	Filling value=0x55 0xa6						
Return Size	2 Bytes	Filling value=0x08						
Year	2 Bytes	0x2006, means the Year 2006						
Month	1 Byte	0x01, means JAN						
Day	1 Byte	0x11, means the Day 11						
Hour	1 Byte	0x12, means twelve o'clock						
Minute	1 Byte	0x22, means 22 minutes						
Week	1 Byte	0x5, means Friday, 0x00 = SUN. ... 0x06 = SAT.						
Time zone	1 Byte	0x1, means GMT-11:00 (Referring to the appendix for time zone value)						

(2)Set the time

It is used for adjusting system date, time, week etc. The format as the following table:

Table2.8.2.1 Command format of setting Time

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	Time	<EOT>
Name	Size	Offset	Description				
<SOH>	1 Byte	0	Start symbol of the command, the value of <SOH>=0x01				
<REV>	1 Byte	1	Fixed, fill in as 'Z'				
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g.: 00 address is showed as 0x30, 0x30.				
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02				
Command Code	1 Byte	5	Command code, the value is= 'E'				
Arg.	1 Byte	6	The value = 'B'				
Time	8 Bytes	7	Time field [2 Bytes] Year E.g. 0x2006 (2006) [1 Bytes] Month E.g. 0x06 (6)				

			[1 Bytes] Day E.g. 0x06 (6) [1 Bytes] Hour E.g. 0x11 (11 o'clock) [1 Bytes] Minute E.g. 0x30 (30) [1 Bytes] Week E.g. 0x03 (TUE.) 0x01 = SUN. ... 0x07 = SAT. [1 Bytes] Time zone E.g. 0x01 (See Appendix 1)
<EOT>	1 Byte	15	Ending symbol of a command 1.If the value=0x04, In-echo 2.If the value =0x03, Echo.

Note of echo: when the <EOT> value= 0x04, no data will be echoed back. When <EOT> value =0x03 and operate successfully, the data is echoed as 'OK'. Status code will appear when the operation fails.

9、Read the basic information of the system

It is used to read back the basic information of the system, for example the GGUU, the version No. etc. The format as the following table:

Table2.9.1 Command Format of Reading the Basic Information of the System

<SOH>	<REV>	Sign Address	<STX>	Command Code	Arg.	<EOT>
Name	Size	Offset	Description			
<SOH>	1 Byte	0	Start symbol of the command, the value of <SOH> =0x01			
<REV>	1 Byte	1	Fixed, fill in as 'Z'			
Sign Address	2 Bytes	2	Unit address of the sign range from 00-99. 00 is broadcast address. E.g: 00 address is showed as 0x30, 0x30.			
<STX>	1 Byte	4	Start symbol of the command <STX> value=0x02			
Command Code	1 Byte	5	Command code, the value ='F'			
Arg.	1 Byte	6	fill in as '0'			
<EOT>	1 Byte	7	The ending symbol of a command Value=0x03, Echo.			

Note of echo: If the command is carried out successfully, the data will be echoed. And its format is as the Table2.9.2.

Table2.9.2 Echoed data format of Reading the System Setting command

SYN head	size	CPU version	TCP/IP Version	File system version	FPGA Version	Length	Width	Type	Rev
Name		Size	Description						
SYN head		2 Bytes	Filling value: 0x55 0xa6						
Size		2 Bytes	Filling value: 0x10						
CPU version		2 Bytes	E.g. 8013 etc.						
TCP/IP version		2 Bytes	Decide according to actual situation						
File system version		2 Bytes	Decide according to actual situation						
FPGA version		2 Bytes	Decide according to actual situation						
Length of the sign		2 Bytes	Decide according to actual situation						
Width of the sign		2 Bytes	Decide according to actual situation						
Type		2 Bytes	Decide according to actual situation						
Rev.		2 Bytes	Decide according to actual situation						

III、Explanation of the 2st communicating

1、Reading data (0x01)

(1) Reading Absolute Address Data(0x0101)

✚ This command is for some special operations. The command format and parameters as the following table:

Table3.1.1.1 Reading Absolute Address Data

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value = 0x01
H	Sub CMD	13	1Byte	Read absolute address, the value = 0x01
I	Arg. Len	14	1 Byte	Parameter of 8 bytes, fill in as 2
J	Flag	15	1 Byte	In the reading data commands, this Flag is invalid
K	Arg.	16	4*2Bytes	[4 Bytes]: Read address [2 Bytes]:Read numbers (not more than 1024 bytes each time) [2 Bytes]: Reserved. Fill in 0
L	Data	24	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.1.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.1.2 Echoed data format after reading absolute address command is performed successfully

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter needed. Fill in as 0
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Byte	No Arg.
L	Data	16	N Bytes	Data read-back

(2) Reading System Files (0x0102)

✚ This command is used to read system files like “CONFIG.SYS”, “SEQUENT.SYS”. The format as the following table:

Table3.1.2.1 Reading System Files

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	02	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1Byte	Read system files. Value= 0x02
I	Arg. Len	14	1 Byte	Need 16 bytes parameter. Fill in as 4

J	Flag	15	1 Byte	In reading data command, this Flag is invalid.
K	Arg.	16	4*4Byte	[12 Bytes]: the file name is “CONFIG.SYS” or “SEQUENT.SYS”, Fill in as 0 behind the name if there are less than 12 Bytes [2 Bytes]: Packet size [2 Bytes]: Packet serial number of reading. (Counted from 1)
L	Data	32	0 Byte	No Data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.2.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.2.2 Echoed data format of reading System Files command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Need 8 bytes parameter. Fill in as 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*1 Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	16	N Bytes	Data read-back

(3) Reading Font Library (0x0103)

✚ This command is used to read the font library. The format as the following table:

Table3.1.3.1 Command format of Reading Font Library

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in detail
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1Byte	Read font files. Value= 0x03
I	Arg. Len	14	1 Byte	Need 16 Bytes parameter. Fill in as 4
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*4Bytes	[12 Bytes]: 12 Bytes for the font file name. The list file name is fixed as FONTLIST.LST. Read one specific font file by read its file name (Different font file names can be found in font list), fill in as 0 behind the name if there are less than 12 Bytes. [2 Bytes]: Package size [2 Bytes]: Packet serial number of reading (Counted from 1)
L	Data	32	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.3.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.3.2 Echoed data format of reading Font Files command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
------	------	--------	------	-------------

A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Need 8 Bytes parameter. Fill in as 2.
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*1 Bytes	[2 Bytes]:File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	20	N Bytes	Reading the data from Font library

(4) Reading Text Files(0x0104)

This command is used to read Text Files in designated section. The format as the following table:

Table3.1.4.1: Reading Text File format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	04	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1Byte	Read Text file. Value= 0x04
I	Arg. Len	14	1 Byte	Need 20 Bytes parameter. Fill in as 5
J	Flag	15	1 Byte	In reading commands, this flag is invalid
K	Arg.	16	4*5 Bytes	[1 Byte]:Disk section code (C、D、E ...) [3 Bytes]:Reserved [12 Bytes]:TEXT FILE LABEL

				[2 Bytes]: Packet size [2 Bytes]: Packet serial number of reading (Counted from 1)
L	Data	36	0 Byte	No data

- ✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.4.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.4.2 Echoed data format of reading Text File command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Need 8 Bytes parameter. Fill in as 2
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*1 Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	20	read size	Text file Data read-back

(5) Reading String File(0x0105)

✚ This command is used to read String Files in a designated section. The format as the following table:

Table3.1.5.1: Command format of Reading String File

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	05	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1Byte	Read String file. Value=0x 05
I	Arg. Len	14	1 Byte	Read 20 Bytes parameter. Fill in as 5
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*5Bytes	[1 Byte]:Drive section code (C、D、E ...) [3 Bytes]:Reserved [12 Bytes]:STRING FILE LABEL [2 Bytes]: Packet size [2 Bytes]: Packet serial number of reading. (Counted from 1)
L	Data	36	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.5.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.5.2 Echoed data format of Reading String File command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	05	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	

C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Read 8 Bytes parameter. Fill in as 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[2 Bytes]: File Size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	20	Read Size	Reading String file Data

(6) Reading Picture File(0x0106)

✚ This command is used to read Picture File in designated section. The format as the following table:

Table3.1.6.1 Read Picture File Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	06	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1 Byte	Read Picture File. Value= 0x 06
I	Arg. Len	14	1 Byte	Read 20 Bytes parameter. Fill in as 5
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*5 Bytes	[1 Byte]: Drive section code (D、E ...) [3 Bytes]:Reserved [12 Bytes]:PICTURE FILE LABEL

				[2 Bytes]: Packet size [2 Bytes]: Packet serial number of reading. (Counted from 1)
L	Data	36	0 Byte	No data

- ✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.6.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.6.2 Echoed data format of reading Picture File command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	06	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Need 8 Bytes parameter. fill in as 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	24	Read size	Reading Picture file Data

(7) Reading Array Picture File(0x0107)

✚ This command is used to read Array Picture files in designated section. The format as the following table:

Table3.1.7.1 Reading Array File Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	07	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Read Data. Value= 0x01
H	Sub CMD	13	1 Byte	Read Array Picture File. Value=0x07
I	Arg. Len	14	1 Byte	Read 20 Bytes parameter. Fill in as 5
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*5Bytes	[1 Byte]:Drive section code (D、E ...) [3 Bytes]:Reserved [12 Bytes]:ARRAY PICTURE FILE LABEL [2 Bytes]: Packet size [2 Bytes]: Packet serial number of reading. (Counted from 1)
L	Data	36	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.7.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.7.2 Echoed data format of read Array File command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	07	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	

C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Need 8 Bytes parameter. Fill in 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	16	Read size	Array file Data read-back

(8) Read Files on Designated Path(0x0108)

🌈 This command is used to read files the names and paths you already know. Command format as following table:

Table3.1.8.1 Reading File on Designated Path Command Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	08	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value= 0x01
H	Sub CMD	13	1 Byte	Read a file on designated path. Value=0x08
I	Arg. Len	14	1 Byte	Arg. Len = ([Arg.]+3)/4
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	N Bytes	[2 Bytes]: Packet size [2 Bytes]: Read serial number packet serial number. (Counted from 1) [Path and File name]:The name and path of the file, E.g. "D:\T\WELCOME.NMG"

				Notes: Path ends with Null (0). File name in upper and lower case differentiated.
L	Data	16+N	0 Byte	No Data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.8.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.8.2 Echoed data format of reading file on designated path command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	08	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	Parameter of 8 Bytes, fill in as 2
I	Arg. Len	14	1 Byte	
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	24	Read size	Reading the content of the file

(9) Reading play Log (0x0109)

✚ This command is used to read play log of system. The format as the following table:

Table3.1.9.1 Format of Read Play Log

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	09	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in
B	Check Sum	2	2 Bytes	

C	Data Len	4	2 Bytes	details
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Read Data. Value= 0x01
H	Sub CMD	13	1 Byte	Read display log. Value= 0x09
I	Arg. Len	14	1 Byte	Need 4 Bytes parameter. fill in as 1
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	4*1 Bytes	[2 Bytes]:Packet size [2 Bytes]: Packet serial number of reading. (Counted from 1)
L	Data	20	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table 3.1.9.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.9.2 Echoed data format of Reading display log command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	09	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	Need 8 Bytes parameter, fill in as2
I	Arg. Len	14	1 Byte	
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[2 Bytes]: File size [2 Bytes]: Packet serial number [4 Bytes]: File size(used for big files)
L	Data	20	N Bytes	Please refer to the table 3.1.9.3 for Log file format in details

Declaration of Log file format

Table3.1.9.3 Log file format declaration

Flag	Counts	Starting time	Read Flag	File counts	Recording location	File path1	File path2	...	Record 1	Record 2	...
Name			Size		Definition						
Flag			2 Bytes		The head mark of the log file, its value="LG"						
Counts			2 Bytes		Record the record numbers in file.						
Starting time			6 Bytes		Time when the log starts (1Byte) Year (1Byte) Month (1Byte) Day (1Byte) Hour (1Byte) Minute (1Byte) Second						
Read flag			1 Byte		If the log file has been read, the value is 1, otherwise it is 0						
File counts			1 Byte		This Byte indicates the item counts of valid File Path.						
Recording location			2 Bytes		Location of current writing.						
File path 1			32 Bytes		It records path of play files.						
File path 2			32 Bytes		It records path of play files.						
File path 3			32 Bytes		It records path of play files.						
...			...		The file paths in total are 128.						
Play record 1			8 Bytes		[1 Byte]: Record the location of the path of the current play file. [1 Byte]: Record play time. 0-23 hours. 1stands for the play record at 1 o'clock. [2 Bytes]: Record date when file was played. The format is: [0 - 4] [5 - 8] [9 - 15] (bit) Day Month Year (starting from the Year 1980, ending by 2107) [2 Bytes]: How many times file was played. [2 Bytes]: Duration of play time (unit: second)						
Play record 2			8 Bytes		[1 Bytes]: Record location of the path of the current play file. [1 Bytes]: Record play time. 0-23 hours. 1stands for 1play record. [2 Bytes]: Record date when file was played. The format is: [0 - 4] [5 - 8] [9 - 15] (bit) Day Month Year (starting from year						

		1980, ending by 2107) [2 Bytes]: How many times file was played. [2 Bytes]: Duration of play time (unit: second)
...	8 Bytes	N th record

Note: If you want to write in the new play file list, please read the play log first, otherwise the previous play log will be replaced.

(10) Reading System Arg. Command (0x010A)

✚ This command is used to read system hardware Arg. Command format as following table:

Table3.1.10.1 Format of Reading System Hardware Arg.

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0A	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Read Data. Value=0x01
H	Sub CMD	13	1 Byte	Read system parameter. Value= 0x0A
I	Arg. Len	14	1 Byte	No parameter needed, fill in as 0
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter.
L	Data	16	0 Byte	No Data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.10.2. If it fails, the status code would be echoed. The status code information in details can be found in Appendix.

Table3.1.10.2 Echoed Data Format of Read System Arg. Command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0A	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
------	------	--------	------	-------------

A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	Need 16 Bytes Arg. fill in as 4
I	Arg. Len	14	1 Byte	
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*4Bytes	[2 Bytes]:CPU Version, like 0x8013 [2 Bytes]: TCP/IP Version number [2 Bytes]: File System Version Number [2 Bytes]: FPGA Version Number [2 Bytes]: Width of LED Sign [2 Bytes]: Height of LED Sign [2 Bytes]: Protocol version [1 Bytes]: Group Address [1 Bytes]: Unit Address
L	Data	32	0 Byte	No Data

(11) Reading Current System Status Command(0x010B)

✚ This command is used to read current system status, like temperature & display status, etc.
The format as the following table:

Table3.1.11.1: Format of Reading System's Current Status

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0B	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value=0x01
H	Sub CMD	13	1 Byte	Read system current status. Value=0x0B
I	Arg. Len	14	1 Byte	No parameter needed.

J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter.
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.11.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.11.2 Echoed data format of Reading current system's status command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0B	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details							
B	Check Sum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte								
H	Sub CMD	13	1 Byte								
I	Arg. Len	14	1 Byte	Need 8 Bytes Arg. Filled in as 2							
J	Flag	15	1 Byte	The value is 0. It means the echoed data.							
K	Arg.	16	4*2 Bytes	[1 Byte]: System status Assignment scheduling status = 0 Emergency message display status = 1 Black screen = 2 Remote control Status = 3 Test mode = 4 Unlimited Connection display mode = 5 Non Word-wrap = 6 [1 Bytes]:In-cabinet temperature (℃) . If there isn't any, it should be 0xff [1 Bytes] :Out-of-cabinet temperature (℃) .If there isn't any, it should be 0xFF [1 Bytes]: Auto ON/OFF. 1=Enable; 0=Disable [1 Bytes]: humidity,value :0 %- 100% [3 Bytes]: Reserved							
L	Data	24	0 Byte	No data							

(12) Reading System SN & MAC Structure (0x010C)

✚ This command is used to read structure such as system SN & MAC addresses. Command format as following table:

Table3.1.12.1 Reading Command Format of System SN & MAC Addresses

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0C	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value=0x01
H	Sub CMD	13	1 Byte	Read system SN & MAC address. Value=0x0C
I	Arg. Len	14	1 Byte	No parameter needed
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No Data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.12.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.12.2 Echoed data format of reading system SN & MAC structure bodies command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0C	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	

F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter Needed
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Bytes	No parameter
L	Data	16	22 Bytes	[12 Bytes]:SerialNo [4 Bytes]:Reserved [6 Bytes]:MAC address

(13) Reading Flash write-in status(Not available now)

✚ This command is used to read status of flash write-in, which is used to judge whether the write-in is successful or not. The format as the following table:

Table3.1.13.1 Reading the format of Flash write-in

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0D	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back. Value=0x01
H	Sub CMD	13	1Byte	Read Flash write-in status, the value=0x0D
I	Arg. Len	14	1 Byte	No parameter needed
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No Data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.13.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.13.2 Echoed format of Reading Flash write-in status command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0D	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	0 Bytes parameter, fill in as 0
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Byte	No parameter
L	Data	16	4 Bytes	[1 Byte]: 0=successful write-in; otherwise, failure [3 Bytes]:Reserved

(14) Reading cabinet working status(0x010E)

- ✚ This command is used to read the cabinet working status and check the working status of the LED sign. The operation of Black screen is needed before this command, otherwise, the status read-back is not accurate. The format as the following table:

Table3.1.14.1 Format of reading cabinet working status

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0E	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value =0x01

H	Sub CMD	13	1 Byte	Read cabinet working status, the value=0x0E
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.14.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.14.2 Echoed Format of reading cabinet working status command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0E	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Byte	No parameter
L	Data	16	400 Bytes	Check data, please refer to Can communication file

NOTE: This command is used only under special circumstances where the control board QS5006 is installed.

(15) Check the CPU update status (0x010F)

✚ This command is for checking whether the CPU updating is successful, if it is successful, the status code is 0x9000, if it is not, the status code is 0x1F01. The format as the following table:

Table3.1.15.1 Format for checking whether CPU updating is successful

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0F	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value =0x01
H	Sub CMD	13	1 Byte	Checking CPU updating status, the value =0x0F
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the CPU updating is successful, the status code is 0x9000. If not, the status code is 0x1F01. If the update operation is not carried out, the status code is 0x1F03, if the updating is undergoing, the status code is 0x1F02.

(16) Read Default Display Style(0x0110)

✚ This command is used to read the system default style, which is for controlling display mode, color and so on (if there is no display mode control in the display file, the default style works. The format as the following table:

Table3.1.16.1 Reading System Default style

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	10	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	

G	Main CMD	12	1 Byte	Data read-back, the value =0x01
H	Sub CMD	13	1Byte	Read default style, the value=0x10
I	Arg. Len	14	1 Byte	No parameter needed.
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter.
L	Data	16	0 Byte	No data.

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.16.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.16.2 Echoed Format of reading System Default Configuration command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	10	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter.
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Bytes	No parameter
L	Data	16	16 Bytes	Structure: Struct { UWORD ID; //55aa UBYTE PlayListLoc;//loc.of play list. UBYTE TimePre0En;//front 0 UBYTE Ddrive; //default disk //default background color UBYTE Dback_color; //default foreground color UBYTE Dfont_color;

				UBYTE Dhor_just; //horiz. alignment UBYTE Dver_just; //verti. alignment UBYTE Dline_space; //line space UBYTE Dfont; //font UBYTED in_mode; // in mode UBYTE Dout_mode; //out mode UBYTE Dspeed; //speed UBYTE Dstay_time; //stay time //auto sentence wrapping UBYTE Dwrap; }DEFAULT_SET Note: PlayListLoc & TimePre0En are available only with firmware of Ver8736, VerA109 or up. Control board QS0925 does not support these two items.
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(17) Reading a File on Designated Path--Extension(0x0111)

 This command is used to read a known file on a known path. It is an extension of the command 0x0108. it can read the file which is larger than 64MB. The format as the following table:

Table3.1.17.1 Format of Reading the Designated File on a Designated Path (Extension)

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	11	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value =0x01
H	Sub CMD	13	1 Byte	Read a designated large file, the value=0x11
I	Arg. Len	14	1 Byte	Arg. Len = ([Arg.]+3)/4
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	N Bytes	[2 Bytes]: Package size. [2 Bytes]: Reserved. [4 Bytes]: Packet serial number of reading (Counted from 1)

				[Path & File]: Path and file name, E.g.: "D:\T\WELCOME.NMG" Note: Path ends with NULL(0), and file name with upper and lower case character differentiation.
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.17.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.17.2 Echoed format of Reading a File on a Designated Path command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	11	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Parameter of 8 Bytes required. Fill in as 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[4 Bytes]:Packet serial number [4 Bytes]:File size
L	Data	24	Read size	To read data contents.

Note: This command is not valid on control board QS0925.

(18) Reading System Information(0x0112)

✚ This command is used to read system information. Details of the information are decided by Firmware. The format as following:

Table3.1.18.1 Format of Reading System Information

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	12	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value =0x01
H	Sub CMD	13	1Byte	Read system parameters, the value= 0x12.
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid.
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.18.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.18.2 Echoed Format of System Information Reading Command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	12	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter, the value=0
J	Flag	15	1 Byte	The value is 0. It means the echoed

				data.
K	Arg.	16	0 Byte	No parameter
L	Data	16	N Bytes	System information data all is in the form of character string.

(19) Reading Error Log(0x0113)

✚ This command is used to read the log data of system errors. The contents of the data are dependent on the FIRMWARE. The format as the following table:

Table3.1.19.1 Format of Reading System Errors

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	13	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Data read-back, the value =0x01
H	Sub CMD	13	1Byte	Read the log data of system errors, the value =0x13
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	In reading data commands, this Flag is invalid
K	Arg.	16	0 Byte	No parameter.
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following Table3.1.19.2. If it fails, the status code would be echoed. The status code information in details can be found in the Appendix.

Table3.1.19.2 Echoed Format of System Error Log Information command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	13	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition

B	Check Sum	2	2 Bytes	in details.
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Byte	No parameter
L	Data	16	N Bytes	Echoed data structure format: [2 Bytes]: ID ID = 0x4552 [2 Bytes]: The starting time.Format is: [0 – 4] [5 - 10] [11 – 15] (bit) Sec*2 Min Hour [2 Bytes]: The starting date. Format is: [0 – 4] [5 - 8] [9 – 15] (bit) Day Month Year (from year 1980 till year 2107) [2 Bytes]: Counts of errors [4 Bytes]: Reserved [2 Bytes]: Error 1 [2 Bytes]: Error 2 Note: The meaning of errors depend on the practical firmware.

Note: This command is not valid on control board QS0925.

2、Information Write-in(0x02)

(1) Absolute address data write-in (0x0201)

✚ This command is used to write in special data. The format as the following table :

Table3.2.1 Command format of writing in absolute address

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1 Byte	Write in Absolute address information, the value=0x01
I	Arg. Len	14	1 Byte	Parameter of 4 Bytes is needed, fill in as 1
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg.	16	4*1 Bytes	[4 Bytes]: Write in address
L	Data	20	Data Size	The data content can not be written in more than 1024 bytes at on time. It is suggested that the content would be 512 bytes or 1024 bytes.

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) System File Write-in(0x0202)

✚ This command is to used write in system files. The format as the following table:

Table3.2.2 Command format of system file write-in

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in system file, the value =0x02
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[12 Bytes]:string, name of system file The name of configuration file is fixed as: CONFIG.SYS The name of list file is fixed as : SEQUENT.SYS The name of Time allocation able is fixed as: RUNTIME.SYS The name of CPU program file is fixed as: PROGRAM.CPU The name of Color Palette file is fixed as: COLORTY.BIN Note: Fill in as 0 if there are not 12 bytes [4 Bytes]: the total size of the file [2 Bytes]:the size of packet (should be the same to all the packet) [2 Bytes]:quantity of packet [2 Bytes]: the current packet. The value of the first packet is 0x01; the value of the second packet is 0x02; the value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number. [2 Bytes]:Note

L	Data	40	Data Size	The data of system file
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- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Font Library File Write-in(0x0203)

- ✚ This command is to used write in font library. The format as the following table

Table3.2.3 Command format of writing in font library file

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in font library file, the value=0x03
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[12 Bytes]: File name The name of the font list is fixed as : FONTLIST. LST. The names of font library are decided by FONTLIST. LST. [4 Bytes]:Total size of the file [2 Bytes]:size of packet (All the packet should be in the same size) [2 Bytes]:quantity of packet [2 Bytes]: the current packet. The value of the first packet is 0x01; the value of the second packet is 0x02; the value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number. [2 Bytes]:Note
L	Data	40	Data Size	Data of font library file

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(4) Text File Write-in (0x0204)

- ✚ This command is used to write Text File into T catalogue under the designated partition. The format as the following table:

Table3.2.4 Text File Command format of writing in Text File

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	04	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1 Byte	Write in Text File, the value=0x04
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[1 Byte]:code of disk partition (D、E ...) [1 Byte]:the ringing times of the buzzer, the maximum is 0x09. when its value=0x00, the buzzer is closed. [12 Bytes]:TEXT FILE LABEL [4 Bytes]:Total size of the file [2 Bytes]:size of packet (All the packets should be the same size.) [2 Bytes]:quantity of packets [2 Bytes]:the current packet. The value of the first packet is 0x01, the value of the second packet is 0x02, the value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number.
L	Data	40	Data Size	Data of Text File

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(5) String File Write-in(0x0205)

- ✚ This command is used to write the String File into S catalogue under the designated partition.
The format as the following table:

Table3.2.5 Command format of writing in String File

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	05	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1 Byte	Write in String File, the value=0x05
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[1 Byte]:Code of disk partition (D、E ...) [1 Bytes]: Reserved, fill in as 0 [12 Bytes]:STRING FILE LABEL [4 Bytes]:Total size of the file [2 Bytes]:Size of packets (All the packets should be the same size.) [2 Bytes]:Quantity of packets [2 Bytes]:The current packet. The value of the first packet is 0x01e; the value of the second packet is 0x02; the value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number.
L	Data	40	Data size	Data of String File

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails.
The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(6) Picture File Write-in(0x0206)

- ✚ This command is used to write in Picture File to P catalogue under the designated partition.
The format as the following table:

Table3.2.6 Picture File command write-in format

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	06	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value= 0x02
H	Sub CMD	13	1 Byte	Write in Picture File, the value= 0x06
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[1 Byte]:Code of disk partition (D, E ...) [1 Bytes]:Reserved, fill in as 0 [12 Bytes]:PICTURE FILE LABEL [4 Bytes]:Total size of file [2 Bytes]:Size of packets (All the packets should be the same size.) [2 Bytes]:Quantity of packets [2 Bytes]:The current packet. The value of the first packet is 0x01.The value of the second packet is 0x02. The value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number.
L	Data	40	Data size	Data of Picture File

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(7) Array Picture File Write-in(0x0207)

✚ This command is to write in Array Picture File to A catalogue under the designated partition. The format as the following table:

Table3.2.7 Command format of writing in Array Picture File

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	07	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1 Byte	Write in Array Picture File, the value= 0x07
I	Arg. Len	14	1 Byte	Parameter of 24 Bytes is needed, fill in as 6
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[1 Byte]:Code of disk partition (D, E ...) [1 Bytes]:Reserved, fill in as 0 [12 Bytes]:ARRAY PICTURE FILE LABEL [4 Bytes]:Total size of the file [2 Bytes]:Size of packets (All the packets should be the same size.) [2 Bytes]:Quantity of packets [2 Bytes]:The current packet. The value of the first packet is 0x01.The value of the second packet is 0x02. The value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number.
L	Data	40	Data size	Data of Array Picture File

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(8) Write a file to a designated path(0x0208)

✚ This command is used to write in a file to a designated path. The format as follows:

Table3.2.8 Command format of writing in a file to a designated path

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	08	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in file, the value=0x08
I	Arg. Len	14	1 Byte	Arg. Len = (Length of Arg. field + 3)/4
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*6 Bytes	[4 Bytes]:Total size of the file [2 Bytes]:Size of packet (All the packets should be the same size.) [2 Bytes]:Quantity of packets [2 Bytes]:The current packet. The value of the first packet is 0x01. the value of the second packet is 0x02.The value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number. [Path and File Name]: Write in the name and the path of the file, E.g.: "D:\T\WELCOME.NMG" Note: the path should end with NULL(0). The file name should be differentiated with upper and lower Case.
L	Data	40	Data Size	Data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(9) Emergency Information Write-in(0x0209)

- ✚ This command is used to write in an emergency message. This message is a Text File.Emergency information means the message is displayed exigently in an active period. The format as follows:

Table3.2.9 Command format of writing in emergency information

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	09	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in emergency information, the value =0x09
I	Arg. Len	14	1 Byte	Parameter of 4 Bytes is needed, fill in as 1
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*1 Bytes	[2 Bytes]:living period is calculated by second. 0 means it displays forever. [1 Bytes]:on & off switch of sound, 1=On, 0 = Off [1 Bytes]:Reserved
L	Data	40	Data size	Data of emergency information(Text file), not larger than 1024 Bytes

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(10) Brightness Control Block Write-in(0x020A)

- The command is used to write in a brightness control block. The display without brightness is useless, E.g. Texcellent. The format as the following table:

Table3.2.10: Command format of writing in Brightness Control block

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	0A	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
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A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1 Byte	Write in brightness information, the value=0x0A
I	Arg. Len	14	1 Byte	Parameter of 12 Bytes is needed, fill in as 3
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*3 Bytes	[2 Bytes]: X place [2 Bytes]: Y place [1 Byte]: Red [1 Byte]: Green [1 Byte]: Blue [2 Bytes]: Width [2 Bytes]: Height [1 Bytes]: Reserved
L	Data	28	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

✚ Note: This command is not valid on control boards QS5003 & QS0925.

(11) Write-in SN&MAC (0x020B)

✚ This command is used to write SN, MAC into the system. The format as the following table:

Table3.2.11: Command format of writing in SN, MAC

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	02	0B	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	

E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in SN, MAC, the value=0x0b
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	0 Bytes	No parameter
L	Data	16	22 Bytes	[12 Bytes] SN [4 Bytes] Reserved [6 Bytes] MAC address

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(12) Write-in Default Display Style(0x010C)

✚ This command is used to write system default display style. This style is used for controlling display modes, colors and so on (If there is no display mode control available in the display files, the default mode will work). The format as the following table:

[Table3.2.12.1](#) Writing of System Default Display Style

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	02	0C	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write in system default Style, the value=0x0c
I	Arg. Len	14	1 Byte	No parameter
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16		No parameter
L	Data	16		Structure Struct {

				UWORD ID; //55aa UWORD Rev; // reserved UBYTE Ddrive; //default disk UBYTE Dback_color; //default background color UBYTE Dfont_color; //default foreground color UBYTE Dhor_just; //horiz. alignment UBYTE Dver_just; //verti. alignment UBYTE Dline_space; //line space UBYTE Dfont; //font UBYTE Din_mode; // in mode UBYTE Dout_mode; //out mode UBYTE Dspeed; //speed UBYTE Dstay_time; //stay time UBYTE Dwrap; //auto sentence wrapping }DEFAULT_SET
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- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(13) Writing a File to a Designated Path—Extension(0x020D)

- ✚ This command is used to write a file to a designated path, especially the file larger than 64MB. The format as the following table:

Table3.2.13 Command Format of Writing a File to a Designated Path

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	02	0D	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Write in information, the value=0x02
H	Sub CMD	13	1Byte	Write a File to a Designated Path, the value is 0x0D
I	Arg. Len	14	1 Byte	Arg. Len = (Length of Arg. field + 3)/4
J	Flag	15	1 Byte	Echo Flag, 1=In-echo; 0=Echo
K	Arg.	16	4*N Bytes	[4 Bytes]: Total size of the file [2 Bytes]: Size of packet (All the packets

				should be the same size.) [4 Bytes]: Quantity of packets [4 Bytes]: The current packet. The value of the first packet is 0x01. the value of the second packet is 0x02. The value of the third packet is 0x03, and so on. Packets should be sent in the order of packet serial number. [Path & File Name]: Write in the name and the path of the file, E.g., "D:\T\WELCOME.NMG" Note: The Path should end with NULL(0)
L	Data	16+N	Data size	Data of writing in files.

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

Note: This command is not valid on control board QS0925.

3、Test Command(0x03)

(1) Connection Test (0x0301)

✚ This command is used to test the connection of the LED sign. If it is connected, some parameters of the sign will return. The format as the following table:

Table3.3.1.1 Command format of Connection Test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command , the value=0x03
H	Sub CMD	13	1 Byte	Connection test command, the value=0x01
I	Arg. Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	No meaning
K	Arg.	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the LED sign is connected, the data will return. The format of these data as the following table:

Table3.3.1.2 Echoed Format of Connection Test Command

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	

G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	The parameter needed 12 Bytes , fill in as 3;
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*3 Bytes	[2 Bytes]: Program version [2 Bytes]: FPGA version [4 Bytes]: IP address, low address is followed by high address, E.g.: 169.254.10.49(A9FE0A31) the return data should be :31 0A FE A9 [2 Bytes]:Sign address, the former 1B is Group, the latter 1B is Unit address [2 Bytes]:Rev
L	Data	28	0 Byte	No data

(2) Auto Test (0x0302)

✚ This command is used to make the LED sign test automatically (The test order: all bright->all Red-> all Green->all Blue-> scan horizontally-> scan vertically>the basic parameter of the sign)

Table3.3.2 Command format of Auto Test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	02	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value=0x03
H	Sub CMD	13	1 Byte	Command for auto test, the value=0x02
I	Arg. Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the Table1.3 for the format of echoed data.

(3) All Bright Test(0x0303)

✚ This command is used to test the LED sign under the condition of Complete Lighting. The format as the following table:

Table3.3.3: Command format of complete lighting test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value=0x03
H	Sub CMD	13	1 Byte	Test command for complete lighting, the value=0x03
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(4) All Red Test(0x0304)

✚ This command is used to test the LED sign under the condition of complete lighting in Red. The format as the following table:

Table3.3.4 Command format of complete lighting in Red test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
------	------	--------	------	-------------

A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value=0x03
H	Sub CMD	13	1 Byte	Test command for complete lighting in Red, the value=0x04
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(5) All Green Test(0x0305)

- ✚ This command is used to test the LED sign under the condition of complete lighting in Green. The format as the following table:

Table3.3.5 Command format of complete lighting in Green test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	05	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name		Offset	Size	Description						
A	SYN Code		0	2 Bytes	Please refer to Table 1.2 for definition in details.						
B	Checksum		2	2 Bytes							
C	Data Len		4	2 Bytes							
D	Source Address		6	2 Bytes							
E	Destination address		8	2 Bytes							
F	Packet Serial		10	2 Bytes							
G	Main CMD		12	1 Byte	Test command, the value=0x03						
H	Sub CMD		13	1 Byte	Test command for complete lighting in Green, the value =0x05						
I	Arg.Len		14	1 Byte	No parameter, fill in as 0						

J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(6) All Blue Test(0x0306)

- This command is to test the LED sign under the condition of complete lighting in Blue. The format as the following table:

Table3.3.6 Command format of complete lighting in Blue test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	06	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value =0x03
H	Sub CMD	13	1 Byte	Test command for complete lighting in Blue, the value=0x06
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(7) Horizontal scanning in All Bright Condition(0x0307)

- The command is used to have the horizontal scanning test on the LED sign in the condition of complete lighting. The format as the following table:

Table3.3.7 Command format of the horizontal scanning test in the condition of complete lighting

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	07	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value =0x03
H	Sub CMD	13	1 Byte	Test command for horizontal scanning in the condition of complete lighting, the value =0x07
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(8) Vertical Scanning in All Bright Condition(0x0308)

✚ This command is to have the vertical scanning test on the LED sign in the condition of complete lighting. The format as the following table:

Table3.3.8: Command format of the vertical scanning test in the condition of complete lighting

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	08	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	

D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value=0x03
H	Sub CMD	13	1 Byte	Test command for vertical scanning in the condition of complete lighting, the value=0x08
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(9) End Test Status (0x0309)

✚ This command is used to end the test on the LED sign. The format as the following table:

Table3.3.9: Command format of ending the test

SYN Code	Check Sum	Data Len	Source Address	Destination Address	Packet Serial	03	09	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Test command, the value=0x03							
H	Sub CMD	13	1 Byte	Test command for ending the test, the value=0x09							
I	Arg.Len	14	1 Byte	No parameter, fill in as 0							
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo							
K	Arg	16	0 Byte	No parameter							
L	Data	16	0 Byte	No data							

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(10) Gray Scale Test (0x030A)

- This command is used to have the gray scale test on the LED sign. The format as the following table:

Table3.3.10: Command Format of Gradual Gray Scale Test

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	03	0A	Arg. Len	Flag	Arg.	Data
I	I	I	I	I	I	I	I	I	I	I	I
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Test command, the value=0x03
H	Sub CMD	13	1 Byte	Test command for gray scale test, the value= 0x0A
I	Arg.Len	14	1 Byte	Parameter of 8 Bytes is needed, fill in as 2.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	8 Bytes	[1 Byte]: Test type 0 = fixed value of gray scale test 1 = gradual gray scale test [1 Byte]: Red portion 1 = with Red portion 0 = without Red portion [1 Byte]: Green portion 1 = with Green portion 0 = without Green portion [1 Byte]:Blue portion 1 = with Blue 0 = without Blue portion [2 Bytes]: Gray scale degree The value of gray scale degree, E.g.: 16 stands for 16 degrees of gray scale. Value is counted from 1

				[2 Bytes]: Rev2
L	Data	24	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

Note: This command is not valid on control board QS0925.

4、Black Screen Command (0x04)

(1) Start the display of black screen (0x0401)

✚ This command is used to make the LED sign be in the status of black screen. The format as the following table:

Table3.4.1 Command format of starting the black screen

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	01	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Black screen command, the value= 0x04
H	Sub CMD	13	1 Byte	Command for starting Black screen, the value=0x01
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) End the Black Screen(0x0402)

✚ This command is used to quit the status of black screen on the LED sign. The format as the following table:

Table3.4.2 Command format of ending the black screen

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	CheckSum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Black screen command, the value= 0x04
H	Sub CMD	13	1 Byte	Command for ending Black screen, the value= 0x02
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Reset the System (0x0400)

✚ This command is used to reset the system of the LED sign. The format as the following table:

Table3.4.3 Command format of resetting the system

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	00	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	CheckSum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Black screen command, the value= 0x04							
H	Sub CMD	13	1 Byte	Command for resetting the system, the value= 0x00							

I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No nada

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(4) Command for Switching Off the LED Sign(0x0403)

- ✚ This order is used to make the LED sign be in a semi-off status. The format as the following table:

Table3.4.4 Command Format of Switching off the LED Sign

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Black screen command, the value=0x04
H	Sub CMD	13	1 Byte	Command for switching off the LED sign, the value=0x03.
I	Arg.Len	14	1 Byte	Parameter of 4 Bytes, fill in as 1
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Whether to show“ USERSHUTDOWNI ” when switching off. 0 = show 1 = do not show [3 Bytes]: Reserved
L	Data	20	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(5) Command for Switching On the LED Sign(0x0404)

✚ This command is used to switch the LED sign from semi-off status to normal display status. The format as the following table:

Table3.4.5 Command Format of Switching On the LED Sign

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Black screen command, the value=0x04							
H	Sub CMD	13	1 Byte	Command for switching the LED sign from semi-off status to normal status, the value=0x04							
I	Arg.Len	14	1 Byte	No parameter, fill in as 0							
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo							
K	Arg	16	0 Byte	No parameter							
L	Data	16	0 Byte	No data							

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(6) Reading Status of the LED sign (0x0405)

✚ This command is used to read the status of the LED sign. It can be known that the sign is on or in semi-off status through this command The format as the following table:

Table3.4.6.1 Command Format of Reading the Status of the LED Sign

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	05	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in							

B	Checksum	2	2 Bytes	details.
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Black screen command, the value=0x04
H	Sub CMD	13	1 Byte	Command for reading the status of the LED sign, the value= 0x05
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data is echoed and its format of communication data is as the following Table 3.4.6.2.

Table3.4.6.2 Echoed data format of reading the status of LED sign command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	04	05	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	0 Bytes	No parameter
L	Data	16	16 Bytes	[1 Byte]: Control board in On/Off status 0 = On 1=Off(“USERSHUTDOWNI” is shown) 2= Off (black screen) [1 Byte]: Driver board in On/Off status 1 = Off, 0 = On [14 Bytes]: reserved

5、Time command (0x05)

(1) Read the time (0x0501)

✚ This command is used to read the time of the LED sign. The format as the following table:

Table3.5.1.1 Command format of reading time

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	05	01	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Time command, the value=0x05
H	Sub CMD	13	1 Byte	Command for reading time, the value=0x01
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Invalid under this command
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the operation is successful, the data is echoed. And its format of communication data is as the following table:

Table3.5.1.1 Echoed data format of reading time command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	05	01	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	

E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Parameter of 8 Bytes, fill in as 2
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4*2 Bytes	[2 Bytes]: Year. E.g.: The Year 2005 will be filled in as 0x2005 [1 Bytes]: Month. E.g.: March. will be filled in as 0x03 [1 Bytes]: Date. E.g.: 22nd will be filled in as 0x22 [1 Bytes]: Hour. E.g.: 15:00 will be filled in as 0x15 [1 Bytes]: Minute. E.g.: 8 minutes will be filled in as 0x08 [1 Bytes]: Week. Value range is [0x00,0x06],E.g. Tuesday will be filled in as 0x02; Sunday as 0x00; Saturday as 0x06 [1 Bytes]: Time-zone. Please refer to the related appendix for more information.
L	Data	24	0 Byte	No data

(2) Adjust Time (0x0502)

🚦 This command is used to adjust the time of the LED sign. The format as the following table:

Table3.5.2 Command format of adjusting time

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	05	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	

G	Main CMD	12	1 Byte	Time command, the value= 0x05
H	Sub CMD	13	1 Byte	Command for adjusting time, the value=0x02
I	Arg.Len	14	1 Byte	Parameter of 8 Bytes, fill in as 2
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*2 Bytes	[2 Bytes]: Year. E.g.: The Year 2005 will be filled in as 0x2005 [1 Bytes]: Month. E.g.: March. will be filled in as 0x03 [1 Bytes]: Date. E.g.: 22nd will be filled in as 0x22 [1 Bytes]: Hour. E.g.: 15:00 will be filled in as 0x15 [1 Bytes]: Minute. E.g.: 8 minutes will be filled in as 0x08 [1 Bytes]: Week. Value range is [0x01,0x07] E.g. Tuesday will be filled in as 0x03; Sunday as 0x01; Saturday as 0x07 [1 Bytes]: Systematical Time-zone. Please refer to the related appendix for more information.
L	Data	32	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Deal with the temperature and humidity(0x0503)

✚ This command is used to deal with the temperature and humidity information. The format as following:

Table3.5.3 Command Format of dealing with temperature and humidity command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	05	03	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	

E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Time command, the value =0x05
H	Sub CMD	13	1 Byte	Command for dealing with the temperature and humidity, the value= 0x03
I	Arg.Len	14	1 Byte	Parameter of 4 Bytes, fill in as 1
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]:Humidity, the value= 0 to 100 [2 Bytes]:Temperature, Temp >= 4; If(Temp & 0x80) Temp = (256 – Temp) 0x80 The unit is (C) ,the highest bit among the 8 bits means minus [1 Bytes]:Brightness
L	Data	32	0 Byte	No data

There is no echoed code after the operation of this command.

(2) Adjust Time -- Extension (0x0504)

This command is used to adjust the time of the LED sign(include second). The format as the following table:

Table3.5.4 Command format of adjusting time

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	05	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	CheckSum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Time command, the value= 0x05
H	Sub CMD	13	1 Byte	Command for adjusting time, the value=0x04
I	Arg.Len	14	1 Byte	Parameter of 12 Bytes, fill in as 3

J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*2 Bytes	[2 Bytes]: Year. E.g.: The Year 2005 will be filled in as 0x2005 [1 Byte]: Month. E.g.: March. will be filled in as 0x03 [1 Byte]: Date. E.g.: 22nd will be filled in as 0x22 [1 Byte]: Hour. E.g.: 15:00 will be filled in as 0x15 [1 Byte]: Minute. E.g.: 8 minutes will be filled in as 0x08 [1 Byte]: Second. E.g.: 8 seconds will be filled in as 0x08 [1 Byte]: Week. Value range is [0x01,0x07] E.g. Tuesday will be filled in as 0x03; Sunday as 0x01; Saturday as 0x07 [1 Byte]: Systematical Time-zone. Please refer to the related appendix for more information. [3 Bytes]: Received.
L	Data	32	0 Byte	No data

If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

6、Play control command (0x06)

(1) Replay the file list(0x0601)

✚ This command is used to replay the file list. The format as the following table:

Table3.6.1 Command format of replaying the file list

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	01	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for replaying the file list, the value= 0X01
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) Replay the current file (0x0602)

✚ This command is used to replay the current file. The format as the following table:

Table3.6.2 Command format of replaying the current file

--

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Play control command, the value=0x06							
H	Sub CMD	13	1 Byte	Command for replaying the current file, the value= 0x02							
I	Arg.Len	14	1 Byte	No parameter, Fill in as 0.							
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo							
K	Arg	16	0 Byte	No parameter							
L	Data	16	0 Byte	No data							

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Pause playing (0x0603)

✚ This command is used to pause playing. The format as the following table:

Table3.6.3 Command format of pausing playing

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Play control command, the value=0x06							

H	Sub CMD	13	1 Byte	Command for pausing playing, the value=0x03
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(4) Continue the play (0x0604)

✚ This command is used to continue the play. The format as the following table:

Table3.6.4 Command format of continuing the play

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for continuing the play, the value=0x04
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(5) Play the next file (Skip the current file) (0x0605)

✚ This command is used to skip the current file and display the next one. The format as the following table:

Table3.6.5 Command format of playing the next file

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	05	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for playing the next file, the value= 0x05
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(6) Play a designated file with priority (Play immediately)(0x0606)

✚ This command is used to play a designated file with priority. After playing the designated file, the system will continue to play the files according to the regular file list. The format as the following table:

Table3.6.6 Command format of playing a designated file with priority

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	06	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	

D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for playing a designated file with priority, the value=0x06
I	Arg.Len	14	1 Byte	Parameter of 16 Bytes, fill in as 4.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*4Bytes	[1 Byte]:Partition number [1 Byte]:Type of the file [12Bytes]:File name (FILE LABEL) [2 Byte]:Reserved
L	Data	32	0 Byte	No data

Note: If the designated file has been saved in the system, the system will pause playing the current file and play the designated file. After playing this designated file, the system will continue to play the files according to the regular file list.

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(7) Read the current file(name, content)(0x0607)

✚ This command is used to read the name or the contents of the current file, E.g. name, contents. The format as following table:

Table3.6.7 Command format of reading the name or the contents of the current file

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	07	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06

H	Sub CMD	13	1 Byte	Command for reading the current file(name, file), the value= 0x07
I	Arg.Len	14	1 Byte	Parameter of 4 Bytes, fill in as 1.
J	Flag	15	1 Byte	neglected under this command.
K	Arg	16	4*1Bytes	[1 Byte]: value 1= Read the file name only Value 2= Read the file name and the content [3 Bytes]:Reserved
L	Data	20	0 Byte	No data

Table3.6.7 Echoed format of reading the current file command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	07	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	Arg. Len= ([Arg.]+3)/4
I	Arg. Len	14	1 Byte	
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	N Bytes	[Path+ file name Bytes]: the path and the file name of the current file
L	Data	16+N	N Bytes	The content of the read-back data

 Note: If there is no current playing file, the echoed code will be 0x6701. This code definition in details can be found in the Appendix.

(8) Read the next file(name, content) (0x0608)

 This command is used to read the name or the contents of the next file. The format as following table:

Table3.6.8 Command format of reading the next file

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	08	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for reading the next file(name, contents), the value=0x08
I	Arg.Len	14	1 Byte	Parameter of 4 Bytes, fill in as 1.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	4*1Bytes	[1 Byte]: Value 1=Read only the name of the file Value 2=Read the name and the content [3 Bytes]: Reserved
L	Data	20	0 Byte	No data

- If the operation is successful, the communication data will be echoed. The data format in details can be found in the Table 3.6.7.
- Note: If there is no current play file, the echoed code will be 0x6702. This code definition in details can be found in the Appendix.

(9) Play the previous file(Skip the current file) (0x0609)

- This command is used to skip the current file and display the previous one. The format as the following table:

Table3.6.9 Command format of playing the previous file

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	09	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in

B	Checksum	2	2 Bytes	details.
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command. Value=0x06
H	Sub CMD	13	1 Byte	Command for playing the previous file, the value=0x09
I	Arg.Len	14	1 Byte	No parameter, fill in as 0
J	Flag	15	1 Byte	Flag of return code, 1=no return; 0=return
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(10) Play forward (0x060a)

✚ This command is used to accelerate playing forward. The format as the following table:

Table3.6.10 Command format of playing forward

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0a	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for playing forward, the value= 0x0a
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter

L	Data	16	0 Byte	No data
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- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(11) Play backward (0x060b)

- ✚ This command is used to play backward. The format as the following table:

Table3.6.11 Command format of playing backward

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0b	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for playing backward, the value= 0x0b
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(12) Play next frame (0x060c)

- ✚ This command is used to control the play of next frame. The format as the following table:

Table3.6.12 Command format of playing next frame

--

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0c	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Play control command, the value=0x06							
H	Sub CMD	13	1 Byte	Command for playing next frame, the value=0x0c							
I	Arg.Len	14	1 Byte	No parameter, fill in as 0.							
J	Flag	15	1 Byte	Echo Flag, 1= In-echo; 0=Echo							
K	Arg	16	0 Byte	No parameter							
L	Data	16	0 Byte	No data							

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(13) Read the current play file(name, content)(0x060D)(extension)

✚ This command is used to read the name or the contents of the current play file, it is the extension command of 0x0607. The format as following table:

Table3.6.13.1 Command format of reading the name or content of the current play file(extension))

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0D	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								

G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for reading the name or content of the current play file, the value=0x0D
I	Arg.Len	14	1 Byte	parameter of 8 Bytes, fill in as 2
J	Flag	15	1 Byte	neglected under this command
K	Arg	16	4*2 Bytes	[1 Byte]: value 1= Read the file name only Value 2= Read the file name and the content [3 Bytes]:Reserved [2 Bytes]:Size of packets [2 Bytes]:Serial number of packets (counted from 1)
L	Data	24	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the following table:

Table3.6.13.2 Echoed format of reading the name or content of current play file command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0D	Arg. Len	Flag	Arg.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	N Bytes	[4 Bytes]:Size of file [2 Bytes]:Serial number of the packets (counted from 1) [2 Bytes]:reserved [path+file name Bytes]:the path and the name of the file
L	Data	16+N	N Bytes	If there is a playing file, the data is the content of the current play file.

Note : if there is no file being displayed, then the return code is: 0x6701, refer to Appendix for

more information.

(14) Read next play file(name or content)(0x060E) (extension)

✚ This command is to read the name or content of the next play file, it is the extension command of 0x0608. The format as the following table:

Table3.6.14 Command format of reading the name or the content of the next play file(extension)

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0E	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06
H	Sub CMD	13	1 Byte	Command for reading the name or content of the next play file, the value=0x0E
I	Arg.Len	14	1 Byte	Parameter of 8 Bytes, fill in as 2.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*2 Bytes	[1 Byte]: value 1= Read the file name only Value 2= Read the file name and the content [3 Bytes]:Reserved [2 Bytes]:Size of packet [2 Bytes]: Serial number of packets (counted from 1)
L	Data	24	0 Byte	No data

✚ If the operation is successful, the data will be echoed. And its format is as the Table3.6.13.2:

✚ Note: If there is no play file, the echoed code will be 0x6702. This code definition in details can be found in the Appendix.

(15) Control the Sound (0x060F)

✚ This command is used to control the buzzer. The format as following table:

Table3.6.15 Command format of controlling the sound

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	06	0F	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Play control command, the value=0x06.
H	Sub CMD	13	1 Byte	Command for controlling the sound, the value=0x0F.
I	Arg.Len	14	1 Byte	Parameter of 8 Bytes, fill in as 2.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*2 Bytes	[1 Byte]: Buzzer switch 1 = buzzer is on; 0 = buzzer is off [1 Byte]: Mode of buzzer. It is valid only when setting is open. 0 = buzzes when new file is received. 1 = buzzes when changing from one file to another. [1 Bytes]: Length of buzzing Values are:'0' to '9' (Second) '0' = mute '1' = buzzes for one second ... [5 Bytes]: Reserved
L	Data	24	0 Byte	No data

✚ This command is valid on the control board QS0925.

7、File control command (0x07)

(1) FDISK partition (0x0701) (No longer supported)

(2) FORMAT (0x0702)

✚ This command is used to format the partition of the disk. The format as the following table:

Table3.6.11.1 Command format of FORMAT command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command, the value= 0x07
H	Sub CMD	13	1 Byte	Command for Formatting the partition, the value=0x02
I	Arg.Len	14	1 Byte	Parameter of 4 Bytes, fill in as 1.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Partition number, 'D', 'E' ... [3 Bytes]: the value is 0
L	Data	20	0 Byte	None

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Creating a new folder (0x0703)

✚ This command is used to create a new folder. The format as the following table:

Table3.7.3 Command format of creating a new folder

--

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	File control command, the value=0x07							
H	Sub CMD	13	1 Byte	Command for creating a new folder, the value= 0x03							
I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4							
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo							
K	Arg	16	N Bytes	[Path of the character string]: To establish the file path. E.g. Establishing folder TEST under disk C: "C:\TEST\ Note: Establishing multilevel folder at one time is not allowed. End with "\"+NULL							
L	Data	16+N	0 Byte	No data							

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(4) RENAME (0x0704)

✚ This command is used to rename a file or a folder. The format as the following table.

Table3.7.4 Command format of renaming a file or folder

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	04	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								

E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command, the value=0x07
H	Sub CMD	13	1 Byte	Command for renaming a file or folder, the value= 0x04
I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	N Bytes	[Path of the character string]: its format as following: Source path & file (folder) name + Space+ Target file (folder) name+ NULL (End with NULL, NULL =0)
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(5) Move a file (0x0705)

✚ This command is used to move the file or the folder. The format as the following table:

Table3.7.5 Command format of moving a file or folder

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	05	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Move files. Value= 0x05
I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	N Bytes	[Size of the character string]: Format as follows:

				Source path & file (folder) name + Space+ Target file (folder) name+ NULL (End by NULL, NULL =0)
L	Data	16+N	0 Byte	No data

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(6) Delete a file(0x0706)

- Under this command, users will be able to delete a file. The format as the following table:

Table3.7.6Command Format of Del

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	06	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value= 0x07
H	Sub CMD	13	1 Byte	Delete file. Value= 0x06
I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	N Bytes	[Path& file name]: Path of the deleted file and the file name (Ended by NULL))
L	Data	16 + N	0 Byte	No data

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(7) Delete all Text File in a designated section(0x0707)

- Under this command, users will be able to delete all the TEXT FILE in a designated partition. The format as the following table:

Table3.7.7 Delete all text file Command Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	07	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Delete the TEXT file in a designated partition. Value=0x07
I	Arg.Len	14	1 Byte	4 Bytes needed. Fill in1.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Partition number, 'D', 'E' ... [3 Bytes]: the value is 0
L	Data	20	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(8) Del all String File in a designated partition. (0x0708)

✚ Under this command, users will be able to delete all the STRING FILE in a designated partition. The format as the following table:

Table3.7.8 Command Format to delete the String files in a designated partition

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	08	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	

E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Delete the SRTING files in a designated partition. Value= 0x08
I	Arg.Len	14	1 Byte	4 Bytes needed. Fill in1.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Partition number, 'D', 'E' ... [3 Bytes]: the value is 0
L	Data	20	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(9) Del all Picture File in a designated partition(0x0709)

✚ Under this command, users will be able to delete all the PICTURE FILE on a designated partition. The format as the following table:

Table3.7.9 Command format of deleting all the PICTURE files in a designated partition

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	09	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Delete the PICTRUE files in a designated partition. Value= 0x09
I	Arg.Len	14	1 Byte	4 Bytes needed. Fill in1.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Partition number, 'D', 'E' ... [3 Bytes]: the value is 0
L	Data	20	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails.

The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(10) Del all Array Picture File files in a designated partition (0x070a)

- Under this command, users will be able to delete the ARRAY PICTURE files in a designated partition. The format as the following table:

Table3.7.10 Command Format to delete all ARRAY PICTRURE files in a designated partition

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0a	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Delete all ARRAY PICTURE file in a designated partition. Value=0x0a
I	Arg.Len	14	1 Byte	4 Bytes needed. Fill in1.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Partition number, 'D', 'E' ... [3 Bytes]: the value is 0
L	Data	20	0 Byte	No data

- If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(11) List the files in the designated folder (0x070b)

- Under this command, the users will be able to list out the file name of the designated folder. The format as the following table:

Table3.7.11.1 DIR Command Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0b	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	File control command. Value=0x07
H	Sub CMD	13	1 Byte	Number of total DIR FILE and files names Value= 0x0b DIR
I	Arg.Len	14	1 Byte	if flag = 0 , Arg. Len= ([Arg.]+3)/4 otherwise , Arg. Len= 0,this filed fill as 0
J	Flag	15	1 Byte	Meanings under this command: If Value=0, use the designated path of the parameter. If Value=1, read the root directory of the default partition If Value=2, read the font library. If Value=3, read the TEXT file folder in the default partition If Value=4, read the STRING file folder in the default partition If Value=5, read the PICTRUE folders in the default partition Value=6, read the ARRAY PICTURE file folder in the default partition
K	Arg	16	N Bytes	File illustration when the Flag =0: [Path length of the folder]: Decided by the length of the character string. End by NULL. Example: "C:\TEST\" is to show the files under the Folder TEST in disk C "C:\FONT\" is to Show the files under FONT folder in disk C. "C:\T\" to show the file under Folder T in disk C.
L	Data	16+N	0 Byte	No data

 Return communication data format

Table3.7.11.2 DIR Command Return Format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0B	Arg. Len	Flag	Flag.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	
I	Arg. Len	14	1 Byte	4 Bytes needed. Fill in1.
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4* 1 Bytes	[2 Bytes]: files under the folder and number of the files [2 Bytes]: reserved
L	Data	20	N Bytes	For data format please see the following information structure.

 File structure:

```
typedef struct //DirectoryEntryStructure
{
    UBYTE bdir_name[11];    // file name
    UBYTE bdir_attr;        // file attribution
    UBYTE bdir_rev;         // reserved
    UBYTE bcr_time_tecth;   // creation time 1. Counted by second
    UWORD wcr_time;         // creation time 2. Counted by hour and minute
    UWORD wcr_date;         // to create a date
    UWORD wlast_acc_time;   // last visit time (hour, minute)
    UWORD wfst_clus_hi;     //
    UWORD wwrt_time;        // last modified time
    UWORD wwrt_date;        // last modified date
    UWORD wfst_clus_lo;
```

```
ULONG dwfile_size;          // file size
}DIRECTORY_ENTRY_STRUCT;
```

(12) List the files in the designated folder (**extension**) (0x070C)

✚ The function of this command is the same as last DIR command, but this command provides the function of sending packets to solve the problem of reading large contents(which includes multi directory entries) , The format as the following table:

Table3.7.12.1 Command format of DIR extension

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0c	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Command of file control, the value is 0x07
H	Sub CMD	13	1 Byte	Command for displaying the total number of files and file names of DIR, the value is 0x0c
I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	The meaning of this command is shown below: When 0, use the designated path in the parameter; When 1, read the root directory in default section When 2, read font library folder; When 3, read TEXT folder in default section; When 4, read STRING folder in default section When 5, read PICTURE folder in default section When 6, read ARRAY PICTURE folder in default section
K	Arg	16	N Bytes	[2 Bytes]: read the number of the directory entries (every directory entry occupies 32 Bytes) [2 Bytes]: start directory entry(means from

				which directory entry it starts reading) If Flag=0, please see the following instruction [Folder path]: is decided by the string length of folder path, ends with NULL. For example: “C:\TEST\” displays the files under TEST folder in Disk C “C:\FONT\” displays the files under FONT folder in Disk C “C:\T\” displays the files under T folder in Disk C
L	Data	16+N	0 Byte	No data

Format of return data:

Table3.7.12.2 Return format of DIR command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	01	0A	Arg. Len	Flag	Flag.	Data
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.
B	Check Sum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	
H	Sub CMD	13	1 Byte	4 Bytes parameter are needed, fill in 1
I	Arg. Len	14	1 Byte	
J	Flag	15	1 Byte	The value is 0. It means the echoed data.
K	Arg.	16	4* 1 Bytes	[2 Bytes]: Numbers of directory entries in the folder [2 Bytes]: reserve
L	Data	20	Data size	Please see the following data structure for data format

File structure:

```
typedef struct //DirectoryEntryStructure
{
    UBYTE bdir_name[11]; //file name
    UBYTE bdir_attr;      //file attribute
    UBYTE bdir_rev;       //reserve
}
```

```

    UBYTE bcr_time_tecth; // time created 1, calculation of second
    UWORD wcr_time;        //time created 2, calculation of hour and minute
    UWORD wcr_date;        //date created
    UWORD wlast_acc_time;   //last access time, hour and minute
    UWORD wfst_clus_hi;     //
    UWORD wwrt_time;        //last modification time
    UWORD wwrt_date;        //last modification date
    UWORD wfst_clus_lo;
    ULONG dwfile_size;      //file size
}DIRECTORY_ENTRY_STRUCT;

```

(13) Obtaining disk information(0x070D)

✚ This command is to read the information of the designated disk, the information includes the type of the disk, the total size and the free size of the disk. The format as the following table:

Table3.7.13.1 Format of command for obtaining disk information

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0D	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Command of file control, the value is 0x07
H	Sub CMD	13	1 Byte	Command of obtaining disk information, the value is 0x0D
I	Arg.Len	14	1 Byte	4 Bytes parameter are needed, so fill in 1
J	Flag	15	1 Byte	This value will be neglected
K	Arg	16	4 Bytes	[2 Bytes]:disk symbol, for example "D:" [2 Bytes]:reserve
L	Data	20	0 Byte	No data

✚ **Format of return data:**

Table3.7.13.2 Return format of command for obtaining disk information

--

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0D	Arg. Len	Flag	Flag.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.3 for definition in details.							
B	Check Sum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte								
H	Sub CMD	13	1 Byte								
I	Arg. Len	14	1 Byte	4 Bytes parameter are needed , so fill in 5							
J	Flag	15	1 Byte	The value is 0. It means the echoed data.							
K	Arg.	16	4*5Bytes	[4 Bytes]:total size [4 Bytes]:available space [12 Bytes]:name of disk							
L	Data	36	0 Byte	No data							

(14) Checking the existence of the designated file (0x070E)

✚ This command is to check the existence of the file in designated path, The format as the following table:

Table3.7.14.1 Command format of checking the existence of designated file

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	07	0E	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L
Item	Name	Offset	Size	Description							
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.							
B	Checksum	2	2 Bytes								
C	Data Len	4	2 Bytes								
D	Source Address	6	2 Bytes								
E	Destination address	8	2 Bytes								
F	Packet Serial	10	2 Bytes								
G	Main CMD	12	1 Byte	Command of file control, the value is 0x07							
H	Sub CMD	13	1 Byte	Check the existence of the file in designated path, the value is 0x0E							

I	Arg.Len	14	1 Byte	Arg. Len= ([Arg.]+3)/4
J	Flag	15	1 Byte	This value will be neglected
K	Arg	16	4*N Bytes	[path]: for example "D:\\T\\A" Note: path is ended with NULL(0)
L	Data	16+N	0 Byte	No data

If the file exists, the return code is 0x9000, if the file doesn't exist, the return code is 0x7E01.

8. Unlimited connection display communication (0x08)

(1) To enter into connection display status(0x0801)

✚ This command serves to start the function of connection display, so that network display in the dot matrix format is possible. The format as the following table:



Table3.8.1 To start the connection display status command format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	08	01	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Unlimited connection display communication, value being 0x08
H	Sub CMD	13	1 Byte	Start connection display status, value being 0x01
I	Arg.Len	14	1 Byte	4 Bytes are needed, so 1 is added.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Direction 0x00 indicates move-left 0x01 indicates move-right [1 Byte]: speed 0x00 fastest ... 0x06 slowest (7 levels) [1 Byte]: means the format of the following data 0 : means the format of the following data is format 1:1 (RGRGRGRG 1Bytes

				array). Note: 8: 8: 8 doesn' t support this format 1 : means sending in default format. When format 1: 1, the sending format is RGRGRGRG ,1Byte arrayed format, Red = 1 bit, Green=1 bit; when format 8:8, the sending format is GR, 2Bytes format, Red = 8 bit, Green = 8 bit; when format 5: 6 : 5 , the sending format is BGR, 2Bytes, Blue = 5 bit, Green = 6 bit, Red = 5 bit; when format 8:8:8, the sending format is BRGN, 4Bytes, Blue=8 bit, Red=8 bit, Green=8 bit, N is empty, which is occupying 8 bit. [1 Byte]: Reserved.
L	Data	20	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) To terminate connection display status(0x0802)

- ✚ This command is used to terminate connection display function. It also can have the system been back to the previous status Schedule Mode status. The format as the following table:

Table3.8.2 Format for terminating connection display status command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	08	02	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Unlimited connection display communication value=0x08
H	Sub CMD	13	1 Byte	Start connection display status value=0x02
I	Arg.Len	14	1 Byte	No parameter needed, so 0 is filled in.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo

K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) To check reception status(0x0803)

- ✚ This command is used to check the reception buffer-status of the connection display. The format as the following table:

Table3.8.3.1 Checking the display status command format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	08	03	Arg. Len	Flag	Arg.	Data
:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	CheckSum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Unlimited connection display communication value=0x08
H	Sub CMD	13	1 Byte	Checking reception status value=0x03
I	Arg.Len	14	1 Byte	No parameter needed, so fill in 0
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

Comparison of return codes

BufPost	No. 1	No. 2	No.3	Return code
0	OLD_BUF	CUR_BUF	DISPLAY_BUF	0x8301
1	CUR_BUF	DISPLAY_BUF	OLD_BUF	0x8302
2	DISPLAY_BUF	OLD_BUF	CUR_BUF	0x8303

Nominal meaning of return codes

The ①, ② and ③ are in the buffer-field: OLD_BUF,CUR_BUF,DISPLAY_BUF

0x8301: buffers ① and ② are being displayed, while buffer ③ is not. Among which ① is moving out, and ② is moving in;

0x8302: buffers ② and ③ is being displayed, while buffer ① is not. Among which ② is moving out, and ③ is moving in.

0x8303: buffers ③ and ① is being displayed, while buffer ② is not. Among which ③ is moving out, and ① is moving in.

If the system is not in the status of unlimited continuous display mode, it will return to 0x8305.

(4) Data download command(0x0804)

✚ This command functions is to download the dot matrix data into the buffer-fields of the LED display. The format as the following table:

Table3.8.4.1 Data download command format

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	08	04	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Unlimited continuous display communication. Value =0x08.
H	Sub CMD	13	1 Byte	Data download command. Value = 0x04.
I	Arg.Len	14	1 Byte	2 is filled in because an 8 Byte parameter is needed.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*2 Bytes	[2 Bytes]: Size of package (All packages should be in the same size.) [2 Bytes]: Total packages [2 Bytes]: Current package. The value of the package 1 is 0x01. The value of package 2 is 0x02. The value of package 3 is 0x03. The other packages are valued in the same manner(Alert: Data sending should be in sequence) [2 Bytes]: Reserved.
L	Data	24	N Bytes	Format of the dot data should be decided by opening the setting of connecting display format. Please find Table 3.8.1.

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

9. Non Word-wrap play(0x09)

(1) Entering into non word-wrap play mode (0x0901)

✚ This command is to activate non word-wrap function. Under non word wrap condition, it will display according to the files in the SEQUENT file. The format as the following table:

Table3.8.1 Command format of activating non word-wrap display mode

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	09	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Non word-wrap display communication. Value=0x09.
H	Sub CMD	13	1 Byte	Start non word-wrap display status. Value=0x01.
I	Arg.Len	14	1 Byte	1 is filled in, because 4 Bytes are needed.
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*1 Bytes	[1 Byte]: Display mode 0x0 = Continuous move 0x1 = Move one frame, then followed by a pause. [1 Byte]: Direction 0x00 Move left 0x01 Move right [1 Byte]: Speed 0x00 Fastest ... 0x06 Slowest (7 levels) [1 Byte]: When in the second display mode, this position is duration time of pause. The unit is "second". Otherwise it carries no meaning.
L	Data	20	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) Terminating non word-wrap display Status (0x0902)

✚ This command functions to close the non word-wrap display function. The format as the following table:

Table3.8.2 Stop command for non word-wrap display status

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	09	02	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	Stop non word-wrap display communication. Value=0x09
H	Sub CMD	13	1 Byte	Stop non word-wrap display. Value=0x02
I	Arg.Len	14	1 Byte	No parameter is needed. Please fill in 0
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	20	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

10、Log-In operation (0x0A)

(1) Log-In operation(0x0A01)

✚ This command is to log in to the system. If Log In is required when the system configuration is being made, the user can operate on the LED sign only through this Log In. The format as the following table:

Table3.10.1 Format of Log In command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	0A	01	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	User management, value =0x0A
H	Sub CMD	13	1 Byte	Log on to the system, value=0x01
I	Arg.Len	14	1 Byte	20 Bytes parameter is needed, so fill in 5
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*5 Bytes	[14 Bytes]:User's name, ends with NULL [6 Bytes]:password
L	Data	36	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(2) Log-Out operation (0x0A02)

✚ This command is to log out the system, once it is logged out, the user can not have any operation then. The format as the following table:

Table3.10.2 Format of Log-Out command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	0A	02	Arg. Len	Flag	Arg.	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	User management, value =0x0A
H	Sub CMD	13	1 Byte	Operation of User's log-out, value=0x02
I	Arg.Len	14	1 Byte	No parameter is needed, so fill in 0
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	0 Byte	No parameter
L	Data	16	0 Byte	No data

✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

(3) Changing password (0x0A03)

✚ This command is to change the User's password, and this operation can be carried out only after Log-on. The format as the following table:

Table3.10.3 Format of changing password command

SYN Code	Check Sum	Data Len	Source Address	Destination address	Packet Serial	0A	03	Arg. Len	Flag	Arg.	Data
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	B	C	D	E	F	G	H	I	J	K	L

Item	Name	Offset	Size	Description
A	SYN Code	0	2 Bytes	Please refer to Table 1.2 for definition in details.
B	Checksum	2	2 Bytes	
C	Data Len	4	2 Bytes	
D	Source Address	6	2 Bytes	
E	Destination address	8	2 Bytes	
F	Packet Serial	10	2 Bytes	
G	Main CMD	12	1 Byte	User management, value =0x0A
H	Sub CMD	13	1 Byte	Changing password, value=0x03
I	Arg.Len	14	1 Byte	28 Bytes parameter is needed, so fill in 7
J	Flag	15	1 Byte	Echoed Flag, 1=In-echo; 0=Echo
K	Arg	16	4*7 Bytes	[14 Bytes]:User's name [6 Bytes]:old password [6 Bytes]:new password

				[2 Bytes]:reserved
L	Data	44	0 Byte	No data

- ✚ If the status code is 0x9000, the command operation is successful. If not, the operation fails. The status code information in details can be found in the Appendix. Please refer to the [Table1.3](#) for the format of echoed data.

Part II: File Format

I 、Text File Format

1、Explanation of the Text File format

Text File is the file which displays the character. It is small in size, and can be sent fast, controlled flexibly, and inserted with Picture File, String File. But it needs to be supported by the font library. The format in the following table:

Table4.1.1 Explanation of the Text File format

Diagram of Text File Format				
Head		Data		EOF
⋮		⋮		⋮
A		B		C
Item	Name	Offset	Size	Description
A	Head	0	7 Bytes	<0x01>Z00<0x02>AX File header marker
B	Data	7	N Bytes	Including display contents and control characters.
C	EOF	8	1 Byte	<0x04> End the file

2、List for Control Character

This character list for controlling the display is applicable for Text File and parts of it also available to the String file and Logo File display format. The one beginning with 0x means HEX. And “”/ ‘’ means ASCII, and the rest means Decimal.

Table4.1.2 Text File/String File/Logo File Controlling Character List

Dec	Hex	Description
1	01	[File Head]: it stands for the start of the file.
2	02	
3	03	[reserved]
4	04	[File end character]: it stands for the end of the file.
5	05	[reserved]
6	06	[Format transfer character] : 1 bytes format This Ctrl Character Valid under the 1st communication of Jetfile II only,it's used to select display protocol.If without this control character, under the 1st communication of Jetfile II , the system will use the ADP2.0 display protocol.Otherwise ,the system will use this table protocol.
7	07	[Flash control character]: 2 Bytes format 0x 07 +'0'= Flash off (Default); 0x 07 +'1'= Flash on

Dec	Hex	Description																																																		
8	08	[Line space control character]: 2Bytes format 0x08 +'0'= line space is 0; 0x08 +'4'= line space is 4; 0x08 +'7'= line space is 7; 0x08 +'1'= line space is 1; 0x08 +'5'= line space is 5; 0x08 +'8'= line space is 8; 0x08 +'2'= line space is 2; 0x08 +'6'= line space is 6; 0x08 +'9'= line space is 9; 0x08 +'3'= line space is 3;																																																		
9	09	[reserved]																																																		
10	0A	[Pattern control character]: 3 Bytes format; 'I'= In pattern, 'O'=Out pattern <table><tr><td>+ ['I']['O'] + 0x 2F =Random</td><td>+ ['I']['O'] + 0x 48 = Fan out</td></tr><tr><td>+ ['I']['O'] + 0x 30 =Jump out;</td><td>+ ['I']['O'] + 0x 49 =Fan in</td></tr><tr><td>+ ['I']['O'] + 0x 31 = Move left;</td><td>+ ['I']['O'] + 0x 4a = Spiral R</td></tr><tr><td>+ ['I']['O'] + 0x 32 =Move right;</td><td>+ ['I']['O'] + 0x 4b = Spiral L</td></tr><tr><td>+ ['I']['O'] + 0x 33 = Scroll left;</td><td>+ ['I']['O'] + 0x 4c = To four corners</td></tr><tr><td>+ ['I']['O'] + 0x 34 = Scroll right;</td><td>+ ['I']['O'] + 0x 4d = From four corners</td></tr><tr><td>+ ['I']['O'] + 0x 35 = Move up;</td><td>+ ['I']['O'] + 0x 4e = To four sides</td></tr><tr><td>+ ['I']['O'] + 0x 36 =Move down;</td><td>+ ['I']['O'] + 0x 4f = From four sides</td></tr><tr><td>+ ['I']['O'] + 0x 37 = Scroll to L/R</td><td>+ ['I']['O'] + 0x 50 = Scroll out from four blocks</td></tr><tr><td>+ ['I']['O'] + 0x 38 = Scroll up</td><td>+ ['I']['O'] + 0x 51 = Scroll in to four blocks</td></tr><tr><td>+ ['I']['O'] + 0x 39 = Scroll down;</td><td>+ ['I']['O'] + 0x 52 =Move out from four blocks</td></tr><tr><td>+ ['I']['O'] + 0x 3a = Fold from L/R</td><td>+ ['I']['O'] + 0x 53 =Move in to four blocks</td></tr><tr><td>+ ['I']['O'] + 0x 3b =Fold from U/D</td><td>+ ['I']['O'] + 0x 54 = Scroll form U/Left, square</td></tr><tr><td>+ ['I']['O'] + 0x 3c =Scroll to U/D</td><td>+ ['I']['O'] + 0x 55 = Scroll from U/right, square</td></tr><tr><td>+ ['I']['O'] + 0x 3d = Shuttle form L/R</td><td>+ ['I']['O'] + 0x 56 =Scroll from D/left, square</td></tr><tr><td>+ ['I']['O'] + 0x 3e =Shuttle from U/D</td><td>+ ['I']['O'] + 0x 57 =Scroll from D/right, square</td></tr><tr><td>+ ['I']['O'] + 0x 3f = Peel off L;</td><td>+ ['I']['O'] + 0x 58 = Scroll from U/left, slanting</td></tr><tr><td>+ ['I']['O'] + 0x 40 =Peel off R;</td><td>+ ['I']['O'] + 0x 59 =Scroll from U/right, slanting</td></tr><tr><td>+ ['I']['O'] + 0x 41 = Shuttle from U/D</td><td>+ ['I']['O'] + 0x 5a =Scroll from D/left, slanting</td></tr><tr><td>+ ['I']['O'] + 0x 42 = Shuttle form L/R</td><td>+ ['I']['O'] + 0x 5b =Scroll from D/right, slanting</td></tr><tr><td>+ ['I']['O'] + 0x 43 = Raindrops;</td><td>+ ['I']['O'] + 0x 5c =Move in from U/left corner</td></tr><tr><td>+ ['I']['O'] + 0x 44 = Random mosaic;</td><td>+ ['I']['O'] + 0x 5d =Move in form U/right corner</td></tr><tr><td>+ ['I']['O'] + 0x 45 = Twinkle stars</td><td>+ ['I']['O'] + 0x 5e =Move in from D/left corner</td></tr><tr><td>+ ['I']['O'] + 0x 46 = Hip-pop</td><td>+ ['I']['O'] + 0x 5f =Move in from D/right corner</td></tr><tr><td>+ ['I']['O'] +0x 47 = Radar scan</td><td>+ ['I']['O'] +0x 60 = Growing up</td></tr></table>	+ ['I']['O'] + 0x 2F =Random	+ ['I']['O'] + 0x 48 = Fan out	+ ['I']['O'] + 0x 30 =Jump out;	+ ['I']['O'] + 0x 49 =Fan in	+ ['I']['O'] + 0x 31 = Move left;	+ ['I']['O'] + 0x 4a = Spiral R	+ ['I']['O'] + 0x 32 =Move right;	+ ['I']['O'] + 0x 4b = Spiral L	+ ['I']['O'] + 0x 33 = Scroll left;	+ ['I']['O'] + 0x 4c = To four corners	+ ['I']['O'] + 0x 34 = Scroll right;	+ ['I']['O'] + 0x 4d = From four corners	+ ['I']['O'] + 0x 35 = Move up;	+ ['I']['O'] + 0x 4e = To four sides	+ ['I']['O'] + 0x 36 =Move down;	+ ['I']['O'] + 0x 4f = From four sides	+ ['I']['O'] + 0x 37 = Scroll to L/R	+ ['I']['O'] + 0x 50 = Scroll out from four blocks	+ ['I']['O'] + 0x 38 = Scroll up	+ ['I']['O'] + 0x 51 = Scroll in to four blocks	+ ['I']['O'] + 0x 39 = Scroll down;	+ ['I']['O'] + 0x 52 =Move out from four blocks	+ ['I']['O'] + 0x 3a = Fold from L/R	+ ['I']['O'] + 0x 53 =Move in to four blocks	+ ['I']['O'] + 0x 3b =Fold from U/D	+ ['I']['O'] + 0x 54 = Scroll form U/Left, square	+ ['I']['O'] + 0x 3c =Scroll to U/D	+ ['I']['O'] + 0x 55 = Scroll from U/right, square	+ ['I']['O'] + 0x 3d = Shuttle form L/R	+ ['I']['O'] + 0x 56 =Scroll from D/left, square	+ ['I']['O'] + 0x 3e =Shuttle from U/D	+ ['I']['O'] + 0x 57 =Scroll from D/right, square	+ ['I']['O'] + 0x 3f = Peel off L;	+ ['I']['O'] + 0x 58 = Scroll from U/left, slanting	+ ['I']['O'] + 0x 40 =Peel off R;	+ ['I']['O'] + 0x 59 =Scroll from U/right, slanting	+ ['I']['O'] + 0x 41 = Shuttle from U/D	+ ['I']['O'] + 0x 5a =Scroll from D/left, slanting	+ ['I']['O'] + 0x 42 = Shuttle form L/R	+ ['I']['O'] + 0x 5b =Scroll from D/right, slanting	+ ['I']['O'] + 0x 43 = Raindrops;	+ ['I']['O'] + 0x 5c =Move in from U/left corner	+ ['I']['O'] + 0x 44 = Random mosaic;	+ ['I']['O'] + 0x 5d =Move in form U/right corner	+ ['I']['O'] + 0x 45 = Twinkle stars	+ ['I']['O'] + 0x 5e =Move in from D/left corner	+ ['I']['O'] + 0x 46 = Hip-pop	+ ['I']['O'] + 0x 5f =Move in from D/right corner	+ ['I']['O'] +0x 47 = Radar scan	+ ['I']['O'] +0x 60 = Growing up
+ ['I']['O'] + 0x 2F =Random	+ ['I']['O'] + 0x 48 = Fan out																																																			
+ ['I']['O'] + 0x 30 =Jump out;	+ ['I']['O'] + 0x 49 =Fan in																																																			
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+ ['I']['O'] + 0x 32 =Move right;	+ ['I']['O'] + 0x 4b = Spiral L																																																			
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+ ['I']['O'] + 0x 35 = Move up;	+ ['I']['O'] + 0x 4e = To four sides																																																			
+ ['I']['O'] + 0x 36 =Move down;	+ ['I']['O'] + 0x 4f = From four sides																																																			
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+ ['I']['O'] + 0x 43 = Raindrops;	+ ['I']['O'] + 0x 5c =Move in from U/left corner																																																			
+ ['I']['O'] + 0x 44 = Random mosaic;	+ ['I']['O'] + 0x 5d =Move in form U/right corner																																																			
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+ ['I']['O'] +0x 47 = Radar scan	+ ['I']['O'] +0x 60 = Growing up																																																			
11	0B	[Special character]: 2 Bytes format <table><tr><td>+ 0x20 MM/DD/YY</td><td>+0x5A With time-zone HH: MIN (-6)</td></tr><tr><td>+ 0x21 DD/MM/YY</td><td>+0x5B With time-zone HH: MIN (-5)</td></tr><tr><td>+ 0x22 MM-DD-YY</td><td>+0x5C With time-zone HH: MIN (-4)</td></tr><tr><td>+ 0x23 DD-MM-YY</td><td>+0x5D With time-zone HH: MIN (-3)</td></tr></table>	+ 0x20 MM/DD/YY	+0x5A With time-zone HH: MIN (-6)	+ 0x21 DD/MM/YY	+0x5B With time-zone HH: MIN (-5)	+ 0x22 MM-DD-YY	+0x5C With time-zone HH: MIN (-4)	+ 0x23 DD-MM-YY	+0x5D With time-zone HH: MIN (-3)																																										
+ 0x20 MM/DD/YY	+0x5A With time-zone HH: MIN (-6)																																																			
+ 0x21 DD/MM/YY	+0x5B With time-zone HH: MIN (-5)																																																			
+ 0x22 MM-DD-YY	+0x5C With time-zone HH: MIN (-4)																																																			
+ 0x23 DD-MM-YY	+0x5D With time-zone HH: MIN (-3)																																																			

Dec	Hex	Description
		+ 0x24 MM.DD.YYYY +0x5E With time-zone HH: MIN (-2) + 0x25 YY +0x5F With time-zone HH: MIN (-1) + 0x26 YYYY +0x60 With time-zone_HH: MIN (+0) + 0x27 MM (number) +0x61 With time-zone HH: MIN (+1) + 0x28 MMM.(char) +0x62 With time-zone HH: MIN (+2) + 0x29 DD (number) +0x63 With time-zone HH: MIN (+3) + 0x2A day of week (number) +0x64 With time-zone HH: MIN (+4) + 0x2B day of week (char) +0x65 With time-zone HH: MIN (+5) + 0x2CHH +0x66 With time-zone HH: MIN (+6) + 0x2DMIN +0x67 With time-zone HH: MIN (+7) + 0x2E SEC +0x68 With time-zone HH: MIN (+8) + 0x2F HH: MIN +0x69 With time-zone HH: MIN (+9) + 0x30 HH: MIN +0x6A With time-zone HH: MIN (+10) (12 hour, AM/PM pre-set) +0x6B With time-zone HH: MIN (+11) +0x31 Temperature +0x6C With time-zone HH: MIN (+12) +0x32 Humidity +0x6D With time-zone HH: MIN (+13) +0x33 Temperature (F) +0x6E With time-zone HH: MIN (-3:30) +0x34 Self-time definition +0x6F With time-zone HH: MIN (+5:30) (Referring to the following explanation) +0x70 With time-zone HH: MIN(+5:45) +0x35 HH (12 Hour) +0x71 With time-zone HH: MIN(+6:30) (... Reserved zone) +0x72 With time-zone HH: MIN(+9:30) +0x53 HH: MIN(offset from GMT) +0x54 With time-zone HH: MIN (-12) +0x55 With time-zone HH: MIN (-11) +0x56 With time-zone HH: MIN (-10) +0x57 With time-zone HH: MIN (-9) +0x58 With time-zone HH: MIN (-8) +0x59 With time-zone HH: MIN (-7) Explanation of the self-time definition:(4 Bytes format) 0x0B + 0x34 + [time of the time zone] + [control character] Time of the time zone (1 Byte): the value ranges from 0x54-0x72. When the value is 0x2f, it stands for inserting the current time of the current time zone Meaning of the control character (1 Byte) : 0 bit: value 0 = 24 hrs value 1 = 12 hrs 1 bit: value 0 = set ":" to flash, value 1 = set ":" to no flash 2 bit: value 0 = without the second graduation, value 1 = with the second graduations. Flashing is invalid when second is on. 3 bit: value 0 = with AM/PM, value 1 =without AM/PM. Under 24 hrs, this feature is invalid; 4,5,6 bit is reserved. 7 bit is1
12	0C	[Frame change]: 1 Bytes format, Start a new page
13	0D	[Line feed]: 1 Bytes format, Start a new line

Dec	Hex	Description
14	0E	[Frame pause time]: Different frames are allowed to have different pause time 0x 0E + '0', Unit of the pause time: Second, Rang '00' – '99'(4 Bytes format) 0x 0E + '1', Unit of the pause time: Millisecond Rang '00'-'99'(4 Bytes format) 0x 0E + '2', Unit of the pause time: Second, Rang '0000'-'9999'(6 Bytes format) 0x 0E + '3', Unit of the pause time: Millisecond, Rang '0000'-'9999'(6 Bytes format) Example: Pause time= 1 second: 0x0E + '0' + '01' Example: Pause time=12 seconds: 0x0E + '0' + '12' Example: Pause time=68 seconds: 0x0E + '0' + '68' Example: Pause time=108 seconds: 0x0E + '2' + '0108' Example: Pause time=4500 seconds: 0x0E + '2' + '4500' Example: Pause time= 50 milliseconds: 0x0E + '1' + '50' Example: Pause time=88 milliseconds: 0x0E + '1' + '88' Example: Pause time=170 milliseconds: 0x0E + '3' + '0170'
15	0F	[Speed]: 2 Bytes format (7 speed levels) + '0': Very Fast; + '1': Fast; + '2': Medium Fast; + '3': Medium; + '4': Medium Slow; + '5': Slow; + '6': Very Slow.
16	10	[reserved]
17	11	[reserved]
18	12	[reserved]
19	13	[String File nest invoke]: 3 Bytes format/6 Bytes format First format :(3 Bytes format) 0x13 + [drive] + [file label] [drive]: Default partition is indicated as" _ ", other partitions are named [file label]: 1 character Second format :(6 Bytes format) 0x13 + 0x0f + [drive] + "S" + [file label] [drive]: Default partition is indicated as" _ ", other partitions are named [file label]: 2 characters Example: Inserting string file A as the file label from disk D: 0x13+'D'+'A' Inserting string file AB as the file label from disk E: 0x13+0x0f + 'E'+'S'+'AB'
20	14	[Picture file nest invoke]: 3 Bytes format format : 0x14 + [drive] + [file label] [drive]: Default partition is indicated as" _ ", other partitions are named [file label]: 1 character Example: Inserting picture file A as the file label from disk D: 0x14+'D'+'A'
21	15	[Whole screen/line background color control character]:5 Bytes format 0x 15 + '-' + BGR (24 Bit) background color of the line. This character should always be put before the 1st displaying character in a line 0x 15 + '+' + BGR (24 Bit) background color of the screen. This character should be put

Dec	Hex	Description
		before the 1 st displaying character in a frame <i>Note: MAIN BOARD QS0925 DOES NOT SUPPROT THIS CONTROL CHARACTER</i>
22	16	[Marquee control character]:3 Bytes format 0x 16 + [mode] + [speed] [Mode]: 1 Byte ‘0’ 3 bits clock wise turning ‘1’ 3 bits anti clockwise turning(not applied now) [Speed]: 1 Byte Reserved. Not available now. <i>Note: MAIN BOARD QS0925 DOES NOT SUPPROT THIS CONTROL CHARACTER</i>
23	17	[Insert Extended ASCII]:2 Bytes format 0x 17 + [Extended ASCII]
24	18	[Protocol-extended character]: N Bytes format The format of the control character: <i>0x 18 + [Length] + [control character]</i>, The word length, which takes 1 Byte, indicates the length of the following control character. 1、 Display Industrial Control Data <i>0x18 + 0x 03 + 0x 01 + [Port] + [Width]</i> The format of displaying the industrial control data The format of the Port and Width are shown by ASCII: E.g. the 5th port is shown as 0x35, the 15th port is shown as 0x3F. The number of Port is from 0 to 15. 2、 Count / Countdown <i>18H+[Length]+[0x0A]+[Figure operation]+[Initial date]+[Basic figure]+[Count tag]+[Reserved]+[Content]</i> <i>[Length]</i>: 1 Byte, indicating the length of the following control characters. Its maximum value is 255. <i>[0x0A]</i> : 1 Byte, fixed value <i>[Figure operation]</i>: 1Byte, indicating the ways of figure operation 0x30 = No operation; 0x31 = Addition; 0x32 = Subtraction; 0x33 = Multiplication; 0x34 = Division; 0x35 = Leaving the remainder <i>[Initial date]</i> : 4 Bytes , Date &Time, indicates the start /end date. Struct <pre>{ UWORD Date; UWORD Time; }</pre> Date(Explanation of the field) - o Bit 0-4: Day of month (1-31) - o Bit 5-8: Month of year (1-12) - o Bit 9-15: Count of years from 1980 (0-127) Time(Explanation of the field) - o Bit 0-4: 2-second count (0-29)

Dec	Hex	Description										
		o Bit 5-10: Minutes (0-59) o Bit 11-15: Hours (0-23) [Basic figure]: 4Bytes, the basic figure value is ten times as the real value. E.g. when the real value is 10, the basic figure value is 100. The value is in the form of ASCII code. E.g. the 1000 is shown as 0x31 0x30 0x30 0x30. The figure operation is only valid for the first parameter. E.g. “hour:%d min:%m”, only the %d value is operated. [Count tag]: 1 Byte , indicating count or countdown. 0x30 means count, and 0x31 countdown. [Reserved]: 3 Bytes , reserved for future use [Content]: display content, consisted of [Time control character] and [Normal display character]. Normal display character means the visible ASCII or Chinese character and so on. The format of Time control character as the following table: <table><tr><th>Control character</th><th>Meaning</th></tr><tr><td>%d</td><td>Day</td></tr><tr><td>%h</td><td>Hour</td></tr><tr><td>%m</td><td>Min.</td></tr><tr><td>%s</td><td>Sec.</td></tr></table> Note: It can be inserted with number [1 to 8] between % and d/h/m/s to show the width. E.g. %8d means the width of the day is 8 characters. If the width is not as long as 8 characters, it is inserted with spaces before the day. %% means the % character. E.g.: Insert “Xx Minutes without accident” to display : The format is : 18H + [Length] + 0x0a + 0x30 + [Date] + [“0000”] + “%m Minutes without accident” Insert “Xx { Days multiply by 3.6} new stores added today” to display: 18H + [length] + 0x0a + 0x33 + [Date] + [“0036”] + “%d new stores added today” 3、Background Picture 18H + [Length] + 0x0B + [Drive] + [File label] [Drive] :Disk name, “@“means the default drive . [File label]: File name of one byte Note: The background picture must be in front of every frame of the visible character. The background picture and the background color can't be set simultaneously. 4、Set the mode of stay time As long as this mode is set, the stay time of file includes the time of in mode. 18H + [0x01] + 0x09 This control character should be in the first frame, which means that the stay time of the file covers the time of display mode processing. 5、Inserting String File of Appointed width 0x18 + 0x18 + 0x08 + [with]+ [alignment]+[path]	Control character	Meaning	%d	Day	%h	Hour	%m	Min.	%s	Sec.
Control character	Meaning											
%d	Day											
%h	Hour											
%m	Min.											
%s	Sec.											

```
+ 'N' (0x4E) 14 * 10 bold English character
+ 'O' (0x4F) 15 * 10 bold English character
+ 'P' (0x50) 16 * 12 bold English character
+ 'Q' (0x51) 24 * 8 bold English character
+ 'R' (0x52) 32 * 8 bold English character
+ 'S' (0x53) 11 * 7 bold English character
+ 'T' (0x54) 12 * 7 bold English character
+ 'U' (0x55) 22 * 12 bold English character
+ 'V' (0x56) 40 * 21 bold English character
+ 'W' (0x57) 24 Chinese Heiti
+ 'X' (0x58) 24 Chinese Xinwei
+ 'Y' (0x59) 24 Chinese Xingkai
+ '`' (0x5a) 24 Chinese Lishu
```

Dec	Hex	Description
		+ '='(0x3d)30*18 standard English character + 'a' (0x5b) 24 Chinese Youyuan + '>'(0x3e)40* 21 standard English character + 'b' (0x5c) 32 Chinese Heiti + '?'(0x3f) 16 * 16 traditional Chinese + 'c' (0x5d) 32 Chinese Xinwei + 'd' (0x5e) 32 Chinese Xingkai + '@'(0x40)7*6 Self-adaptation English character + 'e' (0x5f) 32 Chinese Lishu + 'A'(0x41)14*8 Self-adaptation English character + 'f' (0x60) 32 Chinese Youyuan + 'B'(0x42) 15 * 8 Self-adaptation English character + 'C'(0x43) 16 * 8 Self-adaptation English character + 'g' (0x61) Customer designed character 1 + 'D'(0x44) 24 * 12 Self-adaptation English character + 'h' (0x62) Customer designed character 2 + 'E'(0x45) 32 * 16 Self-adaptation English character + 'i' (0x63) Customer designed character 3 + 'F'(0x46)40 * 21 Self-adaptation English character + 'k' (0x64) Customer designed character 4 + 'G'(0x47) Chinese 40 Songti + 'l' (0x65) Customer designed character 5 + 'H'(0x48) Chinese 40 Heiti + 'm' (0x66) Customer designed character 6 + 'I'(0x49) Chinese 40 Xinwei + 'n' (0x67) Customer designed character 7 + 'J'(0x4A) Chinese 40 Xingkai + 'o' (0x68) Customer designed character 8 + 'K'(0x4B) Chinese 40 Lishu + 'p' (0x69) Customer designed character 9 + 'L'(0x4C) Chinese 40 Youyuan + 'M'(0x4D) Reserve 1 Note: 1. The serial number of the related file can be found in the Font list. 2. There is no special memory treatment for the self- adaptive English character. Users only need to save the related standard English Character
27	1B	[Disposal modes]: 3 Bytes format 0x 1B + '0' + 'a': Stand for constrainedly no-word wrap, no-auto typeset., but auto frame change and continuous left displaying 0x 1B + '0' + 'b': Stands for default disposal mode: word wrap, auto frame change and auto typeset. 0x 1B + '0' + 'c': reserved This control character should be written before any other control character.
28	1C	[Font color]: 2 Bytes format + 0x30 Black + 0x31 Red palette + 0x32 Green palette + 0x33 Yellow palette + 0x34 YGR palette [Character] + 0x35 YGR palette [Horizontal] + 0x36 YGR palette [Wave] + 0x37 YGR palette [Diagonal] + 0x38 BW palette [Vertical gradual change] + 0x39 BW palette [Horizontal gradual change] + 0x3A YW palette [Vertical gradual change]

Dec	Hex	Description
		+ 0x3B YW palette [Horizontal gradual change] + 0x3C RW palette [Vertical gradual change] + 0x3D RW palette [Horizontal gradual change] + 0x3E GW palette [Vertical gradual change] + 0x3F GW palette [Horizontal gradual change] + '/' + BGR (24 Bit) Self-defined color, B = Blue, G = Green, R = Red, which takes 1 Byte
29	1D	[background color of the font]: 2 Bytes format + '0' Black + '1' Red + '2' Green + '3' Yellow + '/' + BGR (24 Bit) Self-defined color, B = Blue, G = Green, R = Red, which takes 1 Byte
30	1E	[Align horizontal]: 2 Bytes format + '0' align center (default) + '1' align left + '2' align right + '3' reserved [Note]: Different lines can have different alignment pattern, but from the last position set, the alignment pattern of the rest lines will be the same,
31	1F	[Align Vertical]: 2 Bytes format + '0' align center + '1' align top + '2' align bottom + '3' reserved
130	82	[1/2 space]

3、Explanation of all the character zones

0x00 – 0x1f [Display control character], Functional control

0x20 – 0x82 [Visual display character], Standard visual English display character

0xa1 – 0xFF [Wide Byte display character], Extended visual display character

II、String File Format

1、Explanation of String File Format

String File can not display by itself, unless it is inserted into Text File. This is why String File shares no File head or End of File. For more about inserted String File, please refer to Test File Format. The following table is about String File Format:

Table5.1.1:String file format

Data ⋮ A				
Item	Name	Offset	Size	Description
A	Data	0	N Bytes	Only the content and control character displayed are available. Refer to Text File Format for control character.

III、Picture File Format

The PICTURE FILE currently supported by JetFileII is a subset of BMP file format from Microsoft Company. And it supports the BMP formats as: single color, 16 colors, 256 colors, 16Bit, 24Bit. And of the 16Bit format BMP, it only supports format 565 with palette. For more detail, please refer to BMP file format from Microsoft Company.

IV、Array Picture File Format

1、Explanation of Array Picture File format

Array Picture file is a file of array format. Compared with Picture File, an Array Picture File can have many frames, while Picture File can only have one; Array Picture File can control every frame's stay time, speed, display mode and so on to your will, while Picture File can only control them to default display configuration. The format in the following table:

Table7.1.1 Format of Array Picture File

Head	Type	Flag	Width	Height	BitPer Point	Total Frame	Data Size	Frame Data	LDW	Rev	Frame Head Struct	Data	Ext Frame Struct
:	:	:	:	:	:	:	:	:	:	:	:	:	:
A	B	C	D	E	F	G	H	I	J	K	L	M	N

Item	Name	Offset	Size	Description
A	Head	0	9 Bytes	<0x01>Z00<0x02>CAPD file heading
B	Type	9	1 Bytes	0 = RGRGRGRG(2bit,1:1 format) 1 = GR(16bit,8:8 format) 2 = BRG(16bit,565 format) 3 = BRG(24bit,8:8:8 format, reserved) 4 = BGRN(32bit,8:8:8:8. Latter 8bit not used) Turn to Table7.1.3 for frame data arrangement
C	Empty Flag	10	1 Bytes	The function works only at non word-wrap play mode. 0: not skip invalid display data in the last frame 1. skip invalid display data in the last frame
D	Width	11	2 Bytes	width of a frame of Array picture
E	Height	13	2 Bytes	height of a frame of Array picture
F	BitPerPoint	15	2 Bytes	Number of bit for one dot. For example, for RG arrays, we use 2 Bits to save one dot. So fill in 2.
G	Total Frame	17	2 Byte	Total frames.
H	Data Size	19	4 Bytes	Total data size (All data in the [Data Field])
I	Frame Data	23	4 Byte	Data size of each frame.
J	LDW Last data width	27	2Byte	Function only at non word-wrap play mode. Record the width of last frame of valid display data. It is reserved at the other modes.
K	Rev	29	2Bytes	Reserved
L	Frame Head Struct1	31	80 Bytes	Control of the 1 st frame to control its mode display, stay time , etc.
...	Frame Head Struct N			control of Nth frame. Number of control framework of frames equals that of frames, i.e. frame heads are as many as frames
M	PDDData1			Array data of first frame, and its size: BitsPerPoint * Width * Height/8(Bytes)
...	PDDData N			array data of Nth frame
N	Ext Frame Struct		4*N	[2 Bytes] : row height [1 Byte] : in mode

				[1 Byte] : out mode If there is one frame that exceeds 16 lines, the control for the exceeding lines is placed here. 4 Bytes for one line.
--	--	--	--	---

Table7.1.2 Format of Frame Head Struct

Head	Rows	Ext_Rows	Row State1	Row State2	...	Row State16	Time Type	Speed	Dir	Stay Time	REV
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		↓
A	B	C	D	E	F	G	H	I	J	K	

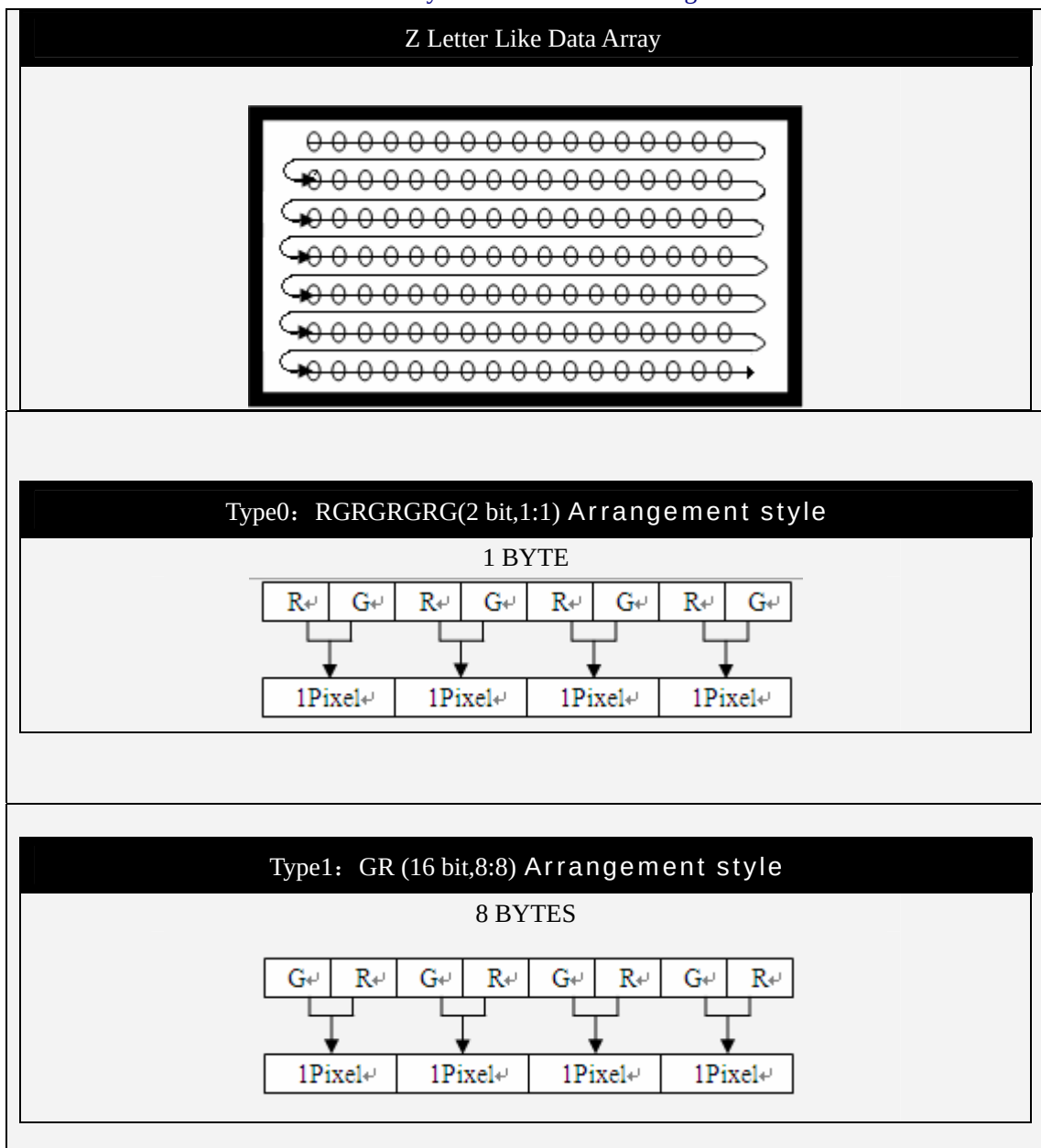
Item	Name	Offset	Size	Description
A	Head	0	2 Bytes	‘FH’ Frame heading mark.
B	Rows	2	1 Bytes	Range: 1- 16. Number of rows in the frame.
C	Ext_Rows	3	1 Bytes	Total lines – Rows. Used when lines are >16.If it is not 0, there are > rows in the frame. Control data for >16 at the very end of the total data.
D	Row State1	4	4 Bytes	Control of the first line [2 Bytes] : row height [1 Byte] : in mode [1 Byte] : out mode
E	Row State2	8	4 Bytes	Control of the second line [2 Bytes] : row height [1 Byte] : in mode [1 Byte] : out mode
F	...	12	4 Bytes	Control of Nth line
G	Row State 16	Rows*4 + 4	4 Byte	Control of the 16 th line [2 Bytes] : row height [1 Byte] : in mode [1 Byte] : out mode
H	Time Type	Rows*4 + 5	1 Byte	Frames stay time. Unit: 0 stands for millisecond 1 stands for second
I	Speed	Rows*4 + 6	1 Byte	‘0’: fastest; ‘6’: slowest (total 7 speeds)
J	Dir	Rows*4 + 8	2 Byte	Valid only in the mode of non word-wrap play. Used to control direction of movement. 0x30 = Move Left 0x31 = Move Right
K	Stay Time	Rows*4	4 Bytes	Stay time. Unit decided by Time Type.

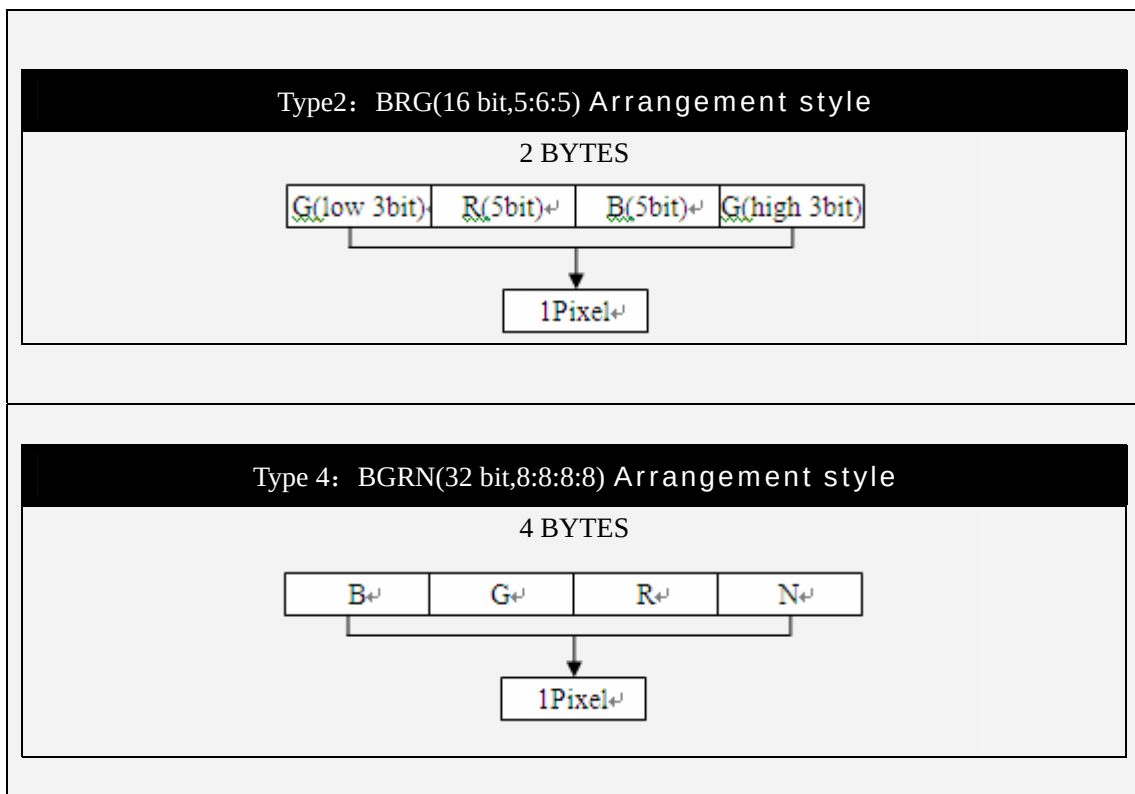
		+ 12		
L	REV	Rows*4 + 16	4 Bytes	Reserved

2、Explanation of Array Picture File data arrangement

Unfolding the array, the data in array picture file is saved in the sequence of letter Z, i.e. at first, data in the first line is saved, then the second and the third and so on, line by line until the last pixel. Data of different types differ in styles of arrangement. Detail as follows:

Table7.1.3 Array Picture File Data Arrangement Format





V、FLW format files(animation format)

1、 Explanation of Animation (FLW) File Format

(1)、 Structure of FLW file

ID: 2BYTE (QS) 51H, 53H	//file marking
FILE_TYPE: 2BYTE	//file type
FRAME_NUM: 2BYTE	//number of frames
WIDTH: 2BYTE	//display width
HEIGH: 2BYTE	//display height
DEPTH: 2BYTE	//display depth
SPEED: 2BYTE	//frame display speed. Unit 1MS
TEXT_FRAME_SIZE: 2BYTE	//not used
REV: 4BYTE	//reserved

Display data

1、 File type

Animation: 00H

Text display array: 01H

Text display symbol(simple character format (4BYTE one character)): 02H

Text display character(complete character format (18BYTE one character)): 03H

2、 Display depth

- 1) 00H: dual color single gray scale. One Byte for 4 dots (GR, GR) Valid only for text format files.
- 2) 01H: dual color 256 gray scale. Two Bytes for 1 dot. (RG)
- 3) 02H: full color16BIT (BGR: 5, 6, 5)
- 4) 03H: full color 256, 3 Bytes for one dot (BGR)
- 5) 04H: full color 256, 4 Bytes for one dot (BGRN)

(2)、 Definitions of Frame Headings

Frameheader=record

```
ID:array[0..1] of Byte;           //Frame starting mark. Fixed at: BK (42H, 4BH)
Frame_size:Longword;             //frame size (including frame head)
Frame_type:word;                 //frame type
Recv1:array[0..3] of Byte;       //reserved
```

end;

1. Frame type

- 1) 00H non-compressed data
- 2) 01H skip row, RLE compressed
- 3) 02H xor,non compressed data
- 4) 03H xor,RLE compressed data
- 5) 04H non compressed data

(3)、 Heading

```
LEN: 2BYTE           //length of data per row
BLOCK_NUM: 2BYTE     //blocks of data per row
```

Skip RLE compressed:

Compressing each row individually. Its definition: number of blocks and value of blocks.

①Number of blocks: WORD type(2 Bytes). Number of blocks in current row.

When D15~D8 is 11H,D7~D0 stand for rows skipped.

②Value of blocks: The top two digits in the first Byte is the block type.

00B: the bottom 6 digits of the first Byte are skipped rows.

01B: The bottom 6 digits of the first Byte are the number of pixels(characters) that need to be moved from the blocks to the images.

After that it's data of N continual pixels.

10B: D5~D0 in the first Byte means the number of pixel(character) that 表示 need to be moved from the blocks to the images.

After that it's the data of that pixel.

VI、 Configuration Files (Config.sys)

The major use of System configuration file is to configure argument of a system. QS5006,QS0712,QS5003,QS0925 configuration file structure as following: :
(change in configuration file structure will not be notified to customers anyway)

```
typedef struct
{
    UWORD    wID;                //0x55aa
    UWORD    wwidth;             //Screen Width
    UWORD    wheight;            //Screen Height
    UBYTE     bpriorityopt;        // 0 = communication as high priority, 0xf0 = display as high priority,
                                   default=communication as high priority.
    UBYTE     blanguagesel;        // 0xf0=traditional Chinese, 0=simple Chinese

    UWORD    wbaud_rate1;         //COM1 baud rate
    UWORD    wbaud_rate2;         //COM2 baud rate
    UWORD    wtcp_time_out;        // TCP time out, in MS
    UBYTE     bDHCPEnable;         //DHCP on/off
    UBYTE     bRev                 //reserved
    UWORD    wcfcard_size;         //cfcard size
    UWORD    wfpga_type;           //fpga type
    UWORD    wfpga_mode;           //fpga mode
    UWORD    wfpga_grade;          //fpga gray scale
}
```

```

UWORD  wfpga_light;           //fpga brightness
UWORD  wfpga_length;          //fpga length
UWORD  wfpga_startline;       //fpga starting line
UBYTE   bsameset;              //same=1,different=0
UBYTE   ledbin;                //

UBYTE   wgroup_addr;           //Group address
UBYTE   wunit_addr;            //Unit address
UBYTE   poweronmsgen;          // 0 = power on message,          0xf0 = no power on message
UBYTE   broadcastipen;         // 0 = address 169.254.10.254, 0xf0 = no response from 169.254.10.254
ULONG   dwip_addr;             //lip_addr

UWORD   wamac_addr[3];         //mac_addr
UWORD   SoftSecVer;            //software section version number

UWORD   wsoftver;              //software version number
UWORD   whardver;              //hardware version number

POWER_TIME_STRUCT  spower_time;          //power on/off timer

UBYTE   btime_zone;             //time zone, please turn to appendix 1
UBYTE   bremote;                //0: remote control,  1: no remote control
UBYTE   bplay_mode;             //0: instant processing download file,  1: download file processing delayed
UBYTE   bdaylight ;             //daylight saving time

POWER_TIME_STRUCT  shalfight_time;       // half-lit time

UBYTE   Systemstat;             // set the starting state of a system, value=0 means schedule mode
UBYTE   uselogin;               //log in or not, default=no log in;
UBYTE   bslavenum;              //number of slave board
UBYTE   TempFlag ;              // Flag=0, temperature in Celsius Centigrade; flag=1, temperature Fahrenheit

UWORD   NewLineWidth;           // width of a new line
UWORD   NewLineHeight;          // height of a new line
UWORD   Offline_X;              //X
UWORD   Offline_Y;              //Y

UBYTE   TempValue;              // value of temperature shift

UBYTE   SerialNo[13];           // serial number
UBYTE   DateTime[8];            // date/time for renewing files
UBYTE   LedName[11];            //Led name
UBYTE   BrightCtrl[18];         // brightness control, BrightCtrl[0] grade options, grade 1-15, 0=automatic, and
the latter 17 grades are cross tabulation for brightness

```

```

    UBYTE Rev[64];          // reserved
}CONFIG_HEAD_STRUCT;

typedef struct
{
    UBYTE      bis_user;      //when the value is true, it stands for valid framework
    UBYTE      boff_hour;
    UBYTE      boff_min;
    UBYTE      bon_hour;
    UBYTE      bon_min;
    UBYTE      barev[3];
}POWER_TIME_STRUCT;

```

VII、Play List (SEQUENT.SYS)format

SEQUENT.SYS is used for controlling the display sequence of the files. The file name is fixed as SEQUENT.SYS. The format as the following table:

Table 10.1.1 SEQUENT File

Name	Offset	Size	Value	Description
Marking	0	2Bytes	'SQ'	Mark of the file
File type	2	1Byte		Mark of the type , Value is :<0x04> schedule all files in all partitions
valid marking	3	1Byte	0	0 stands for SEQUENT FILE valid 1 stands for SEQUENT FILE invalid
File amount	4	2Bytes		Stand for the total amount of the scheduled files. It is used for determining the amount of the schedule structure of the following related files
Reserved	6	2Bytes	0	Reserved
Data content	8			Structure (36Bytes): typedef struct { //Partition number UBYTE bdrive; //File types: TEXT is abbreviated to T, ARRAY PICTURE is abbreviated to A UBYTE btype; //File name(FILE LABEL),if this field =<0x0f> , it's mean use //bfile_name field for file name UBYTE bfile_label; }

				//Week repetition. Please refer to the explanation of week repetition UBYTE bweek_rep; //Start date & time. Construct :DATE_TIME_STRUCT DATE_TIME_STRUCT sbegin_data_time; //End date & time DATE_TIME_STRUCT send_data_time; //Sum of verify(the content of the whole file) UWORD dwcheck_sum; //size of the file UWORD dwfile_size; //File name 8.3 format UBYTE bafilename[12]; }FULL_SEQUENT_FILE_STRUCT; Note: when the sum of verify is the same as that of the size of the file, system will consider the two to be the same file. [Explanation of week repetition] D7 D6 D5 D4 D3 D2 D1 D0 8bit in total , Dx stands for the sequence(No.1, No 2...) within bweek_rep D0 : Stands for Sunday. True time: During the period "sbegin_date - send_date", this file will be played every Sunday; False time: the file will not be played on every Sunday D1 : Stands for Monday. True time: During the period "sbegin_date - send_date", this file will be played every Monday; False time: the file will not be played on every Monday D2 : Stands for Tuesday. True time :During the period "sbegin_date - send_date", this file will be played every Tuesday; False time: the file will not be played on every Tuesday D3 : Stands for Wednesday. True time :During the period "sbegin_date - send_date", this file will be played every Wednesday; False time: the file will not be played on every Wednesday D4 : Stands for Thursday. True time :During the period "sbegin_date - send_date", this file will be played every Thursday; False time: the file will not be played on every Thursday D5 : Stands for Friday. True time :During the period "sbegin_date - send_date", this file will be played every Friday; False time: the file will not be played on every Friday.
--	--	--	--	--

				D6 : Stands for Saturday. True time :During the period "sbegin_date - send_date", this file will be played every Saturday; False time: the file will not be played on every Saturday. D7 : True time: Neglect all the setting(including week, date, time). The file will be played forever. [Construct: DATE_TIME_STRUCT] <pre>typedef struct { //Year. 2000 Year is represent by the number 0x2000(BCD) UWORD wyear; //Month. January to December is represented by the number 0x01-0x12. Other numeral will be defaulted as January(BCD) UBYTE bmonth; //Date , Date 1-31 will be represent by the number 0x01-0x31. Other numeral will be defaulted as 1st Date.(BCD) UBYTE bday; //Hour. Hour 0-23 will be represented by the number 0x0-0x23. Other numeral will be defaulted as the 1st hour.(BCD) UBYTE bhour; //Minute, Minute 0-59 will be represent by the number 0x00-0x59. Other numeral will be defaulted as 1st minute.(BCD) UBYTE bminute; //When true, date is neglected. Otherwise date is valid UBYTE bdate_use_flag; //When true, time is neglected, otherwise time is valid UBYTE btime_use_flag; }DATE_TIME_STRUCT;</pre>
--	--	--	--	---

VIII、 LOGO File Format

This file is saved at "C:\\\\LOGO.SYS". It is used to save Logo files. The format is as follows:

Table 12.1.1 Logo file format

Name	Offset	Size	Value	Description
File flag	0	4 Bytes	'LOGO'	File header
Length	4	2 Bytes		Logo length. Unit in pixel.

Width	6	2 Bytes		Logo width. Unit in pixel.
X coordinates	8	2 Bytes		Display x & y coordinates of logo. Upper left corner is the origin.
Y coordinates	10	2 Bytes		
Valid flag	12	1 Byte	0x01/0x00	Whether this flag is valid or not. 1 means valid. 0 means invalid)
Dynamic updating flag	13	1 Byte	0x01/0x00	If this flag is not set(0), logo will not be updated dynamically. If time or other special characters or symbols are inserted, please set to 1
Transparent	14	1 Byte	0x01/0x00	1 = transparent, 0 = not transparent
Reserved	15	1 Byte		Reserved
[data]	16	N Bytes		Logo data including display character and control character. For control character, refer to Table4.1.2.
File end flag		1 Byte	0x04	Its value is 0x04

Appendix I Correspondence Relations of Time Zones and Values

+0x00	With time zone HH: MIN	(-12)
+0x01	With time zone HH: MIN	(-11)
+0x02	With time zone HH: MIN	(-10)
+0x03	With time zone HH: MIN	(-9)
+0x04	With time zone HH: MIN	(-8)
+0x05	With time zone HH: MIN	(-7)
+0x06	With time zone HH: MIN	(-6)
+0x07	With time zone HH: MIN	(-5)
+0x08	With time zone HH: MIN	(-4)
+0x09	With time zone HH: MIN	(-3)
+0x0a	With time zone HH: MIN	(-2)
+0x0b	With time zone HH: MIN	(-1)
+0x0c	With time zone HH: MIN	(+0)
+0x0d	With time zone HH: MIN	(+1)
+0x0e	With time zone HH: MIN	(+2)
+0x0f	With time zone HH: MIN	(+3)
+0x10	With time zone HH: MIN	(+4)
+0x11	With time zone HH: MIN	(+5)
+0x12	With time zone HH: MIN	(+6)
+0x13	With time zone HH: MIN	(+7)
+0x14	With time zone HH: MIN	(+8)
+0x15	With time zone HH: MIN	(+9)
+0x16	With time zone HH: MIN	(+10)
+0x17	With time zone HH: MIN	(+11)
+0x18	With time zone HH: MIN	(+12)
+0x19	With time zone HH: MIN	(+13)
+0x1a	With time zone HH: MIN	(-3:30)
+0x1b	With time zone HH: MIN	(+5:30)
+0x1c	With time zone HH: MIN	(+5:45)
+0x1d	With time zone HH: MIN	(+6:30)
+0x1e	With time zone HH: MIN	(+9:30)

Appendix II Version Compatibility Test Command and Return

Format

Send font string:

“QS:VER”, which is composed of 6 fonts

Return format:

QS: + [CPU version, 2 Byte] + [FPGA version, 2 Byte] + AD: + [IP address, 4 Byte]

Appendix III Second Category Command Quick Check List

Major Cat. Minor Cat.	0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08	0x09	0x0A
0x01	0101	0201	0301	0401	0501	0601	0701	0801	0901	0A01
0x02	0102	0202	0302	0402	0502	0602	0702	0802	0902	0A02
0x03	0103	0203	0303	0403	0503	0603	0703	0803		0A03
0x04	0104	0204	0304	0404		0604	0704	0804		
0x05	0105	0205	0305	0405		0605	0705			
0x06	0106	0206	0306			0606	0706			
0x07	0107	0207	0307			0607	0707			
0x08	0108	0208	0308			0608	0708			
0x09	0109	0209	0309			0609	0709			
0x0a	010A	020A	030A			060A	070A			
0x0b	010B	020B				060B	070B			
0x0c	010C	020C				060C	070C			
0x0d	010D	020D				060D	070D			
0x0e	010E					060E	070E			
0x0f	010F					060F				
0x10	0110									
0x11	0111									
0x12	0112									
0x13	0113									
0x00				0400						

Definition of Commands

Major Category	Command	Definition of Command
Read Data	0101	Read absolute address
	0102	Read system configuration files
	0103	Read font library files
	0104	Read TEXT FILE
	0105	Read STRING FILE
	0106	Read PICTURE FILE
	0107	Read ARRAY PICTURE FILE
	0108	Read files on designated path
	0109	Read display log
	010A	Read system parameters
	010B	Read current status
	010C	Read system SN,MAC structure body
	010D	Read FLASE Write-in status

	010E	Read cabinet working status
	010F	Check CPU update status
	0110	Read back the default setting (Display mode)
	0111	Read back the designated path File--Extended
	0112	Read back the system information
	0113	Read back the information of wrong log
Write Data	0201	Write absolute address data
	0202	Write system files.
	0203	Write font library files
	0204	Write Text File
	0205	Write String File
	0206	Write Picture File
	0207	Write Array Picture File
	0208	Write files on designated path
	0209	Write emergency messages
	020A	Write brightness control module
	020B	SN,MAC structure body write-in
	020C	Write-in default setting (Display mode)
	020D	Write the file to the designated path—extended
Test command	0301	Connect for testing
	0302	Auto test display
	0303	All bright display
	0304	All red display
	0305	All green display
	0306	All blue display
	0307	All bright scan display
	0308	All bright vertical scan display
	0309	End testing status
	030A	Test of gray degree
Black screen display command	0401	Start black screen display
	0402	End black screen display
	0403	Close the command of the LED screen
	0404	Open the command of the LED screen
	0400	System reset
Time command	0501	Read time command
	0502	(Write) time adjustment command
	0503	Command of dealing with temperature and humidity
Display control	0601	Re-display file list
	0602	Re-display current file.
	0603	Display pause
	0604	Continue displaying
	0605	Display next file (Skip current file)

	0606	Priority display of a file (Display immediately)
	0607	Read current display file (Name, content)
	0608	Read next display file (Name, content)
	0609	Display previous file (Skip current file)
	060a	Fast forward
	060b	Backward
	060c	Next frame
	060d	Read the current display file (name and content) (extended)
	060e	Read next display file (name and content) (extended)
	060F	Command for controlling sound
File Control	0701	FDISK
	0702	FORMAT
	0703	CRTDIR create a folder
	0704	RENAME
	0705	MOVE
	0706	DEL
	0707	Delete a designated section Text file
	0708	Delete a designated section String file
	0709	Delete a designated section Picture file
	070a	Delete a designated section Array Picture File file
	070b	DIR total files and file name display command
	070c	Command of displaying the total number of DIR files and the file names
	070d	Command of obtaining disk information
	070e	Check the existence of the designated file
	0700	Perform default initialization on disk
Unlimited connection display communication	0801	Start connection display mode
	0802	End connection display mode
	0803	Check reception status
	0804	Data download command
Non word-wrap display status	0901	Open non word-wrap display status
	0902	Close non word-wrap display status
Log-on operation	0A01	Log-on operation
	0A02	Lot-out operation
	0A03	Changing password

Appendix IV Valid File Labels

A File Label is a single ASCII font. Messages are stored in or retrieved from the memory file that is defined by this label in the Memory Configuration.

File Labels can be anywhere in the range 20H through 7EH inclusive.

The only special case occurs when File Label “0”(30H) is used for a priority TEXT file (see “Priority TEXT files”) which is pre-configured as a set portion of memory outside of the Memory Pool.

Table 28: Valid File Labels

	30H-‘0’	40H-‘@’	50H-‘P’	60H-‘`’	70H-‘p’
21H-‘!’	31H-‘1’	41H-‘A’	51H-‘Q’	61H-‘a’	71H-‘q’
22H-‘”’	32H-‘2’	42H-‘B’	52H-‘R’	62H-‘b’	72H-‘r’
23H-‘#’	33H-‘3’	43H-‘C’	53H-‘S’	63H-‘c’	73H-‘s’
24H-‘\$’	34H-‘4’	44H-‘D’	54H-‘T’	64H-‘d’	74H-‘t’
25H-‘%’	35H-‘5’	45H-‘E’	55H-‘U’	65H-‘e’	75H-‘u’
26H-‘&’	36H-‘6’	46H-‘F’	56H-‘V’	66H-‘f’	76H-‘v’
27H-‘”’	37H-‘7’	47H-‘G’	57H-‘W’	67H-‘g’	77H-‘w’
28H-‘(’	38H-‘8’	48H-‘H’	58H-‘X’	68H-‘h’	78H-‘x’
29H-‘)’	39H-‘9’	49H-‘I’	59H-‘Y’	69H-‘i’	79H-‘y’
2AH-‘*’		4AH-‘J’	5AH-‘Z’	6AH-‘j’	7AH-‘z’
2BH-‘+’	3BH-‘;’	4BH-‘K’	5BH-‘[’	6BH-‘k’	7BH-‘{’
2CH-‘,’	3CH-‘<’	4CH-‘L’		6CH-‘l’	7CH-‘ ’
2DH-‘-’	3DH-‘=’	4DH-‘M’	5DH-‘]’	6DH-‘m’	7DH-‘}’
	3EH-‘>’	4EH-‘N’	5EH-‘^’	6EH-‘n’	
2FH-‘/’	3FH-‘?’	4FH-‘O’	5FH-‘_’	6FH-‘o’	

Notes:

1. The blanks of in the above table mean that the character can not be used as a file name.
2. ADP only supports file names with length 1—2
3. Jetfile II(Chapter II) supports file names with length up to 8.3(8 Byte main file name and 3 Byte extension name). File names are comprised with characters in the above table.

Appendix V Status Codes and Their Meanings

Status Codes	Phenomena	Meanings
0x9000 0x4B4F	If the data is processed correctly but the command is not to read messages, it will return the value 0x9000.	Data received correctly and is being processed correctly.
0x9002	Sum check error	Sum check is incorrect
0x9003		Address is wrong is not sent back.
0x9004	Major category command invalid.	Major category command in the command categories invalid.
0x9005	Minor category command invalid.	Minor category command in the

		command categories invalid.
0x9006	Package length incorrect.	Returns when the package data length plus the package length exceeds the actual received length.
0x9008	File not found	File does not exist when operating on files.
0x9009	File received till end	Package sequential number too big to read.
0x9010	File opening failure	File not found or file system is erroneous.
0x9011	Minor category command is not supported in this system.	Commands of this category are not supported in this system.
0x9012	When the file is being written in	Maybe the left space of the disk is not enough
0x9013	When the file is being written in	When the file is being written in, the size of the package is bigger than 1500 or the size of data is bigger than the package size
0x9014	When the file is being written in	When the file is written in, the data packages are not sent in sequence
0x9015	when the file is being deleted	When the file is being deleted, the file is open
0x9030	Please log on first, if password management is pre-set, it will return when the operation is being done without logging on	log on the operation
0x9031	the password is wrong	
0x9032	The user's name is wrong	
0x9033	The old password is not correct	
0x9000	The change of password is successful	
0x9035	It is logged on in elsewhere	
0x1101	Over volume in reading absolute addresses.	Incorrect or invalid parameters in commands.
0x1102	Error in reading or writing hard disk.	May be incorrect reading of addresses.
0x1F01	When the CPU is being updated	CPU update unsuccessful
0x1F02	When the CPU is being updated	CPU is being updated
0x1F03	Check the update status of CPU	Check the update status of CPU, but the system, didn't do the update
0x2000	Management RAM application failure in writing files.	Management RAM application failure.

0x2101	Over size data in writing in.	Writing file size can not exceed 320K at a time.
0x2102	When the file is being written in, there is no enough space in the disk	The file written in is bigger than the space of disk
0x2103	When the file is being written in	When the file is written into disk C, the left space is not enough and it returns
0x2104	When the file is being written in	When the file is written into disk D, the left space is not enough and it returns
0x2105	When the file is being written in	When the file is written into disk E, the left space is not enough and it returns
0x2106	When the file is being written in	When the file is written into disk F, the left space is not enough and it returns
0x2901	Message oversize.	Emergency message can not exceed 1024 Bytes.
0x3A01	Gray scale test	When the system does no support the Gray scale test
0x5201	Time setting failure.	Time setting failure.
0x6701	No current display file.	When reading current or the next display file or content, if there is no file for display in the system, the system will return to previous stage.
0x6702	Failure in opening the current display file	Failure in opening the current display file
0x6703	When the current display file is being read, it is too big	The current display file is too big, the current command can not read it, please use Extended Read-back command.
0x7201	when the disk is being formatted	Failure in formatting the disk
0x7301	Failure in creating a file.	Failure in creating a file.
0x7401	Failure in changing the files' name	Failure in changing the files' name
0x7402	The path format is not correct when changing the name	The path format is not correct when changing the name, please refer to protocol
0x7501	Failure in moving a file.	Failure in moving a file.

0x7601	Failure in deleting a file.	Failure in deleting a file.
0x7B01	Failure in opening the file	Failure in opening the file
0x7D01	when the disk information is being read	Failure in reading the disk information
0x7E01	Check whether the designated file exists or not	It returns when the designated file doesn't exist
0x8301	Check unlimited connection display status.	① & ② buffer are being displayed; ③ buffer is not being displayed. ① is moving out and ② is moving in.
0x8302	Check unlimited connection display status.	② & ③ buffer are being displayed; ① buffer is not being displayed. ② is moving out and ③ is moving in.
0x8303	Check unlimited connection display status.	③ & ① buffer are being displayed; ② buffer is not being displayed. ③ is moving out and ① is moving in.
0x8305	Check unlimited connection display status.	The system is not in the unlimited connection status.
0x8306	The data input is out of bound, the receiving buffer exceeds	When the data is bigger than the receiving buffer
0x8307	The format of the sending data is not correct. The LED screen is at 8:8:8:n but the data sent in appears in the format of 1:1	8:8:8 doesn't support 1:1 format

The figure on the thousandth digit represents major category of error; the figure on the hundredth digit represents minor category of error. Any code starting by 9 represents common error.

Appendix VI Explanation of Terms

Default disk:

When writing ADP command, if there is no designated disk, write the file into the default disk. If the default disk is set to be "E", send the command :<0x01>Z00<0x02>AADisplay Test<0x03> then the file will be written into disk E, for more detail, please refer to the first communication .

Text File

Controlling the display text file with JetFileII display protocol, the feature is the text format is saved as character coding, and it occupies little space, but it needs the support of font base to display correctly. It is saved under the contents of T in the data disk.

String file

String file is to be inserted into the Text File and displayed. The string file itself can not

display. The saving format is the same as Text, and it is saved under S contents. .



picture file

JetFileII supports RG tri-color, 16 colors, 256 colors, 16 bit and 24 bit BMP format. .



array picture

Same as BMP type, but the saved data is related to the type of LED screen. Refer to File Format for more details. .



fonts,

It keeps the shape of the characters for displaying Text File.



configuration

It is used to save the setting of the system.



Play List

Control the display order, display time and date of the display files.



Default display parameters

It is the value the system gives when no display parameter is set for the display file. For example, the in-mode of a file is not set, the system will give out a default parameter as the in-mode.



FONTLIST.LST

It is used to record the number of fonts and the property of fonts. .

Appendix VII Sample Function

Check Sum Function

```
//*****
//**** Name:MsgCountChecksumTwo
//**** Func:check sum
//**** Arg.:
//****      buf          - Data Buffer
//****      begin        - starting loction
//****      end           - Ending loction
//**** Return:
//****      Check sum
//*****
ULONG MsgCountChecksumTwo(UBYTE *buf, ULONG begin, ULONG end)
{
    ULONG i, check_sum;

    check_sum = 0;
    if (end >= begin)
    {
        i = end - begin;
        buf += begin;
        while(i--)
        {
            check_sum += *buf++;
        }
    }
    return check_sum;
}
```